UNIVERSITY OF KELANIYA / SENG 34213
FOREVER HOTEL MANAGEMENT SYSTEM

Test Case Register

Consolidated Test Case Evidence | 307 Recorded Cases

Reporting date: 8 November 2026. Evidence is identified by subsystem repository, branch, commit SHA, and GitHub Actions run date; comparisons across separate repository commits do not constitute a single deployed six-subsystem release test.

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1. Register overview and status definitions

This register consolidates the requirement-traced test cases across all six Forever Hotel subsystems as of 8 November 2026, preserving test identifiers, descriptions, requirements, test levels, preconditions, inputs/steps, expected and actual results, status labels, and source paths. Reported PASS (CI suite) indicates that the containing automated test suite passed in GitHub Actions CI; live cross-service staging, physical hardware, and manual client sign-off items remain explicitly marked as NOT TESTED or NOT VERIFIED.

Subsystem	Recorded cases	Suite-backed passing cases	Unverified / not tested
HW	33	28	5
MAD	58	53	5
FDS	43	34	9
FOSS	54	43	11
KMS	82	76	6
WKMS	37	30	7
All six	307	264	43

FOSS entries are updated to the authoritative 54-case register (TC-FOSS-001 to TC-FOSS-054) committed in forever-hotel/food-ordering/documents/testing/test-register-source.csv (commit 9ac1e6f / develop a69ba0a16c). HW entries reflect the 8 November 2026 develop CI suites (5e3cd6bf32 / 440761c34a) and QA deliverable package (63673ba704). FDS retains all 43 original IDs alongside its 27 October 2026 local verification addendum and 8 November 2026 develop CI (4b5ef57d4c).

2. Hotel Website (HW)

Repository: forever-hotel/hotel-website | Branch: develop (merged from pre-develop 440761c34a; QA package: 63673ba704) | Tested revision: 5e3cd6bf323bc005e9efb86b5e09ba0e3015e727 | CI date: 8 November 2026

Evidence: https://github.com/forever-hotel/hotel-website/actions/runs/42504096480,
https://github.com/forever-hotel/hotel-website/actions/runs/42504096452,
https://github.com/forever-hotel/hotel-website/actions/runs/42504621927,
https://github.com/forever-hotel/hotel-website/actions/runs/42504621987,
https://github.com/forever-hotel/hotel-website/actions/runs/42500427127

Register entries: 33; CI-suite-linked pass: 28; not tested/not verified: 5. Automated CI executes 460 backend unit tests, 79 PostgreSQL integration tests (7 skipped), 58 frontend Node tests, 3 browser E2E journeys, 48 browser quality checks, 22 staging tooling tests, and 8 load-runner fixture tests.

TC-HW-001 | PASS (CI suite)

only currently valid offers appear, including start and excluding end

Requirement: HW-01 | Level: Frontend Node test | Priority: P1

Preconditions	Prepared isolated fixtures
Input / steps	Run named test with fixture data
Expected result	Named assertions succeed
Recorded actual result	Test suite passed in matched GitHub Actions CI; individual assertion transcript not archived.
Test file / evidence	frontend/tests/promotions.test.mjs
Additional note	Client-facing offers rules, not live Manager promotion publication.

TC-HW-002 | PASS (CI suite)

empty and malformed responses produce an empty list

Requirement: HW-01 | Level: Frontend Node test | Priority: P1

Preconditions	Prepared isolated fixtures
Input / steps	Run named test with fixture data
Expected result	Named assertions succeed
Recorded actual result	Test suite passed in matched GitHub Actions CI; individual assertion transcript not archived.
Test file / evidence	frontend/tests/promotions.test.mjs

TC-HW-003 | NOT TESTED

Approved hotel background, policies and map content verified by hotel representative

Requirement: HW-02 | Level: Manual acceptance | Priority: P1

Preconditions	Client-approved content and browsers
Input / steps	Review public information
Expected result	Accurate location, policies and hotel information
Recorded actual result	No verified complete execution or acceptance evidence at this commit.
Test file / evidence	frontend/src/features/home/content.ts
Additional note	Static content paths exist; client sign-off not evidenced.

TC-HW-004 | PASS (CI suite)

should return public catalogue fields given room types exist

Requirement: HW-03 | Level: Unit | Priority: P1

Preconditions	Prepared isolated fixtures
Input / steps	Run named test with fixture data
Expected result	Named assertions succeed
Recorded actual result	Test suite passed in matched GitHub Actions CI; individual assertion transcript not archived.
Test file / evidence	backend/src/room-types/room-types.service.spec.ts

TC-HW-005 | PASS (CI suite)

catalogue parses API records and formats minor units as LKR

Requirement: HW-03 | Level: Frontend Node test | Priority: P1

Preconditions	Prepared isolated fixtures
Input / steps	Run named test with fixture data
Expected result	Named assertions succeed
Recorded actual result	Test suite passed in matched GitHub Actions CI; individual assertion transcript not archived.
Test file / evidence	frontend/tests/rooms.test.mjs

TC-HW-006 | PASS (CI suite)

malformed catalogue records are not shown as bookable rooms

Requirement: HW-03 | Level: Frontend Node test | Priority: P1

Preconditions	Prepared isolated fixtures
Input / steps	Run named test with fixture data
Expected result	Named assertions succeed
Recorded actual result	Test suite passed in matched GitHub Actions CI; individual assertion transcript not archived.
Test file / evidence	frontend/tests/rooms.test.mjs

TC-HW-007 | PASS (CI suite)

room comparison supports two and three selections without duplicates

Requirement: HW-04 | Level: Frontend Node test | Priority: P1

Preconditions	Prepared isolated fixtures
Input / steps	Run named test with fixture data
Expected result	Named assertions succeed
Recorded actual result	Test suite passed in matched GitHub Actions CI; individual assertion transcript not archived.
Test file / evidence	frontend/tests/comparison.test.mjs

TC-HW-008 | PASS (CI suite)

a fourth room is blocked until an existing room is removed

Requirement: HW-04 | Level: Frontend Node test | Priority: P1

Preconditions	Prepared isolated fixtures
Input / steps	Run named test with fixture data
Expected result	Named assertions succeed
Recorded actual result	Test suite passed in matched GitHub Actions CI; individual assertion transcript not archived.
Test file / evidence	frontend/tests/comparison.test.mjs

TC-HW-009 | PASS (CI suite)

respects maintenance, active statuses, unassigned inventory and checkout boundaries

Requirement: HW-05 | Level: PostgreSQL integration | Priority: P1

Preconditions	Prepared isolated fixtures
Input / steps	Run named test with fixture data
Expected result	Named assertions succeed
Recorded actual result	Test suite passed in matched GitHub Actions CI; individual assertion transcript not archived.
Test file / evidence	backend/test/integration/availability.integration-spec.ts

TC-HW-010 | PASS (CI suite)

stay validation rejects invalid dates, ranges and occupancy

Requirement: HW-05 | Level: Frontend Node test | Priority: P1

Preconditions	Prepared isolated fixtures
Input / steps	Run named test with fixture data
Expected result	Named assertions succeed
Recorded actual result	Test suite passed in matched GitHub Actions CI; individual assertion transcript not archived.
Test file / evidence	frontend/tests/availability.test.mjs

TC-HW-011 | PASS (CI suite)

returns 503 rather than false availability on inventory failure

Requirement: HW-05 | Level: Unit | Priority: P1

Preconditions	Prepared isolated fixtures
Input / steps	Run named test with fixture data
Expected result	Named assertions succeed
Recorded actual result	Test suite passed in matched GitHub Actions CI; individual assertion transcript not archived.
Test file / evidence	backend/src/availability/availability.controller.spec.ts

TC-HW-012 | PASS (CI suite)

should return 201 with the public guest fields and store a bcrypt hash given valid details

Requirement: HW-06 | Level: PostgreSQL integration | Priority: P1

Preconditions	Prepared isolated fixtures
Input / steps	Run named test with fixture data
Expected result	Named assertions succeed
Recorded actual result	Test suite passed in matched GitHub Actions CI; individual assertion transcript not archived.
Test file / evidence	backend/test/integration/guest-registration.integration-spec.ts

TC-HW-013 | PASS (CI suite)

should return 409 EMAIL_ALREADY_REGISTERED given the email is used in a different case

Requirement: HW-06 | Level: PostgreSQL integration | Priority: P1

Preconditions	Prepared isolated fixtures
Input / steps	Run named test with fixture data
Expected result	Named assertions succeed
Recorded actual result	Test suite passed in matched GitHub Actions CI; individual assertion transcript not archived.
Test file / evidence	backend/test/integration/guest-registration.integration-spec.ts

TC-HW-014 | PASS (CI suite)

should return 400 given an unknown field such as a role

Requirement: HW-06 | Level: PostgreSQL integration | Priority: P1

Preconditions	Prepared isolated fixtures
Input / steps	Run named test with fixture data
Expected result	Named assertions succeed
Recorded actual result	Test suite passed in matched GitHub Actions CI; individual assertion transcript not archived.
Test file / evidence	backend/test/integration/guest-registration.integration-spec.ts

TC-HW-015 | PASS (CI suite)

should issue a JWT and return the authenticated profile given valid credentials

Requirement: HW-07 | Level: PostgreSQL integration | Priority: P1

Preconditions	Prepared isolated fixtures
Input / steps	Run named test with fixture data
Expected result	Named assertions succeed
Recorded actual result	Test suite passed in matched GitHub Actions CI; individual assertion transcript not archived.
Test file / evidence	backend/test/integration/guest-auth.integration-spec.ts

TC-HW-016 | PASS (CI suite)

should return the same unauthorized error given an unknown email or wrong password

Requirement: HW-07 | Level: PostgreSQL integration | Priority: P1

Preconditions	Prepared isolated fixtures
Input / steps	Run named test with fixture data
Expected result	Named assertions succeed
Recorded actual result	Test suite passed in matched GitHub Actions CI; individual assertion transcript not archived.
Test file / evidence	backend/test/integration/guest-auth.integration-spec.ts

TC-HW-017 | PASS (CI suite)

should reject an expired token

Requirement: HW-07 | Level: Unit | Priority: P1

Preconditions	Prepared isolated fixtures
Input / steps	Run named test with fixture data
Expected result	Named assertions succeed
Recorded actual result	Test suite passed in matched GitHub Actions CI; individual assertion transcript not archived.
Test file / evidence	backend/src/common/utils/jwt-token.service.spec.ts

TC-HW-018 | PASS (CI suite)

Guest completes new website reservation and receives a reference

Requirement: HW-08 | Level: Cross-service E2E | Priority: P1

Preconditions	Authenticated guest, signed quote fixture, and disposable PostgreSQL / browser E2E runner
Input / steps	Complete online reservation and fixture payment flow via backend unit/integration and frontend E2E suite
Expected result	Reservation created with reference and confirmed status in My Bookings without overbooking
Recorded actual result	Passed in Backend CI (Run 42504096480) and Frontend Browser E2E CI (Run 42504096452 / 42504621987).
Test file / evidence	backend/src/bookings/bookings.service.spec.ts; frontend/tests/booking.test.mjs; frontend/tests/booking.browser.mjs
Additional note	Verified within isolated CI/fixture payment boundary; live external PayHere merchant account settlement remains separate.

TC-HW-019 | PASS (CI suite)

Concurrent final-room booking never double-books inventory

Requirement: HW-09 | Level: Cross-service concurrency | Priority: P1

Preconditions	Disposable CI PostgreSQL database with one available physical room and two competing guest transactions
Input / steps	Execute concurrent reservation transactions in booking-race.integration-spec.ts and load runner fixture
Expected result	Exactly one persisted booking winner and one ROOM_UNAVAILABLE conflict; zero overbooking
Recorded actual result	Passed in Backend PostgreSQL Integration CI (Run 42504096480 / 42504621927) and performance/test/runner.test.mjs.
Test file / evidence	backend/test/integration/booking-race.integration-spec.ts; performance/test/runner.test.mjs
Additional note	Verified on disposable CI PostgreSQL; 300-user staging load contention remains unexecuted.

TC-HW-020 | NOT TESTED

Promotion code applied to checkout with authoritative eligibility

Requirement: HW-10 | Level: Cross-service E2E | Priority: P1

Preconditions	Approved MAD promotion and live HW checkout redemption integration
Input / steps	Select public offer handoff in UI and submit unsupported promo fields to backend contract test
Expected result	Public offer handoff/removal works in UI and unsupported checkout promo fields are rejected; live MAD redemption pending
Recorded actual result	UI offer handoff (booking.browser.mjs) and unsupported promo rejection (http-consumer-contracts.integration-spec.ts) pass in CI; live MAD checkout redemption is not implemented.
Test file / evidence	frontend/tests/offers.test.mjs; frontend/tests/booking.browser.mjs; backend/test/integration/http-consumer-contracts.integration-spec.ts
Additional note	Retained as NOT TESTED for live cross-service MAD promotion redemption.

TC-HW-021 | NOT TESTED

Online payment provider checkout, callback and receipt reconciliation

Requirement: HW-11 | Level: Provider integration | Priority: P1

Preconditions	Approved live/sandbox PayHere merchant account and exposed webhook callback
Input / steps	Verify signed PayHere webhook MAC/amount/status in unit/E2E suites vs live external PayHere checkout
Expected result	Tampered/unpaid redirects cannot confirm booking; valid signed notification reconciles payment
Recorded actual result	PayHere signature verification, failure isolation, and forged-return rejection pass in CI (payhere-provider.spec.ts, payhere-payments.service.spec.ts, payment.test.mjs, smoke.test.mjs); live external PayHere callback not executed.
Test file / evidence	backend/src/payments/payhere-provider.spec.ts; backend/src/payments/payhere-payments.service.spec.ts; frontend/tests/payment.test.mjs
Additional note	Retained as NOT TESTED for live external PayHere merchant gateway settlement.

TC-HW-022 | NOT TESTED

Booking confirmation email delivered after paid booking

Requirement: HW-12 | Level: Provider integration | Priority: P1

Preconditions	Integrated payment confirmation and live Brevo email provider account
Input / steps	Trigger confirmation email dispatch, replay, and failure handling
Expected result	Idempotent receipt dispatch without duplicate sends; payment remains valid on email provider failure
Recorded actual result	Unit idempotency and sandbox outcome handling pass in confirmation-email.service.spec.ts (Run 42504096480); live external guest inbox delivery not executed.
Test file / evidence	backend/src/payments/confirmation-email.service.spec.ts
Additional note	Retained as NOT TESTED for live external email inbox delivery.

TC-HW-023 | PASS (CI suite)

Guest account profile and upcoming/past booking list accepted

Requirement: HW-13 | Level: Browser acceptance | Priority: P1

Preconditions	Authenticated guest and seeded upcoming, past, and cancelled bookings
Input / steps	Run dashboard unit tests, cross-guest HTTP contract tests, and browser E2E guest dashboard journey
Expected result	Only owned upcoming/past/cancelled bookings shown; foreign guest reads/mutations rejected with 404
Recorded actual result	Passed in Frontend CI (dashboard.test.mjs, booking.browser.mjs Run 42504096452) and Backend PostgreSQL CI (http-consumer-contracts.integration-spec.ts Run 42504096480).
Test file / evidence	frontend/tests/dashboard.test.mjs; frontend/tests/booking.browser.mjs; backend/test/integration/http-consumer-contracts.integration-spec.ts
Additional note	Cross-guest isolation and browser dashboard pagination/mobile/sign-out verified in CI.

TC-HW-024 | PASS (CI suite)

Guest modifies or cancels confirmed booking subject to policy

Requirement: HW-14 | Level: Cross-service E2E | Priority: P1

Preconditions	Confirmed booking, frozen policy clock, and signed change preview
Input / steps	Modify or cancel booking before/after cutoff or with stale revision in backend unit, HTTP integration, and frontend management tests
Expected result	Eligible owned revisions succeed; cutoff/stale revisions rejected; retry preserves attempt key
Recorded actual result	Passed in Backend CI (booking-management.service.spec.ts, http-consumer-contracts.integration-spec.ts Run 42504096480) and Frontend CI (management.test.mjs Run 42504096452).
Test file / evidence	backend/src/bookings/booking-management.service.spec.ts; frontend/tests/management.test.mjs; backend/test/integration/http-consumer-contracts.integration-spec.ts
Additional note	Policy enforcement verified in CI; live bank refund settlement remains a manual staff action.

TC-HW-025 | NOT TESTED

Services and facilities public page accessibility/client acceptance

Requirement: HW-15 | Level: Manual acceptance | Priority: P1

Preconditions	Browser/accessibility review
Input / steps	Inspect service page at mobile/desktop resolutions
Expected result	Approved services are accurate and accessible
Recorded actual result	No verified complete execution or acceptance evidence at this commit.
Test file / evidence	frontend/src/app/services/page.tsx

TC-HW-026 | PASS (CI suite)

sends plain-text email to configured management with guest reply-to

Requirement: HW-16 | Level: Unit / mocked provider | Priority: P1

Preconditions	Prepared isolated fixtures
Input / steps	Run named test with fixture data
Expected result	Named assertions succeed
Recorded actual result	Test suite passed in matched GitHub Actions CI; individual assertion transcript not archived.
Test file / evidence	backend/src/contact/contact.service.spec.ts
Additional note	Unit verifies dispatch contract, not real delivered email.

TC-HW-027 | PASS (CI suite)

contact feedback distinguishes provider acceptance, sandbox and failures

Requirement: HW-16 | Level: Frontend Node test | Priority: P1

Preconditions	Prepared isolated fixtures
Input / steps	Run named test with fixture data
Expected result	Named assertions succeed
Recorded actual result	Test suite passed in matched GitHub Actions CI; individual assertion transcript not archived.
Test file / evidence	frontend/tests/contact.test.mjs

TC-HW-028 | PASS (CI suite)

limits each connection IP and expires the window

Requirement: HW-16 | Level: Unit | Priority: P1

Preconditions	Prepared isolated fixtures
Input / steps	Run named test with fixture data
Expected result	Named assertions succeed
Recorded actual result	Test suite passed in matched GitHub Actions CI; individual assertion transcript not archived.
Test file / evidence	backend/src/contact/contact-rate-limit.guard.spec.ts

TC-HW-029 | PASS (CI suite)

Responsive and accessible hotel website on mobile/tablet/desktop

Requirement: HW-17 | Level: Browser acceptance | Priority: P1

Preconditions	Production Next.js build and headless browser quality runner across 320px, 375px, 768px, 1024px, and 1440px viewports
Input / steps	Execute npm run test:quality (48 responsive, accessibility, keyboard navigation, and 320px booking checks)
Expected result	Zero horizontal overflow or audit failures across all 5 viewports; keyboard forms/calendar/FAQ/comparison pass
Recorded actual result	All 48 browser quality and accessibility checks passed in Frontend CI (Run 42504096452 / 42504621987).
Test file / evidence	frontend/tests/booking.browser.mjs
Additional note	Automated browser viewport and accessibility checks pass in CI; final client visual sign-off remains separate.

TC-HW-030 | PASS (CI suite)

should attach verified claims given a valid bearer token

Requirement: NFR-12 | Level: Unit | Priority: P1

Preconditions	Prepared isolated fixtures
Input / steps	Run named test with fixture data
Expected result	Named assertions succeed
Recorded actual result	Test suite passed in matched GitHub Actions CI; individual assertion transcript not archived.
Test file / evidence	backend/src/common/guards/jwt-auth.guard.spec.ts

TC-HW-031 | PASS (CI suite)

should reject a signature with the correct length but incorrect value

Requirement: NFR-12 | Level: Unit | Priority: P1

Preconditions	Prepared isolated fixtures
Input / steps	Run named test with fixture data
Expected result	Named assertions succeed
Recorded actual result	Test suite passed in matched GitHub Actions CI; individual assertion transcript not archived.
Test file / evidence	backend/src/common/utils/jwt-token.service.spec.ts

TC-HW-032 | PASS (CI suite)

loads without schema synchronization and round-trips relationships, money, dates and payment references

Requirement: NFR-16 | Level: PostgreSQL integration | Priority: P1

Preconditions	Prepared isolated fixtures
Input / steps	Run named test with fixture data
Expected result	Named assertions succeed
Recorded actual result	Test suite passed in matched GitHub Actions CI; individual assertion transcript not archived.
Test file / evidence	backend/test/integration/booking-payment.integration-spec.ts

TC-HW-033 | PASS (CI suite)

Container CI verifies readyness under DB outage

Requirement: NFR-19 | Level: Docker CI | Priority: P1

Preconditions	Disposable docker-compose stack in GitHub Actions
Input / steps	Stop test PostgreSQL; request backend /health
Expected result	503 during DB outage; recovery once DB restored
Recorded actual result	Test suite passed in matched GitHub Actions CI; individual assertion transcript not archived.
Test file / evidence	docker-compose.yml
Additional note	This row maps to completed container CI step; source is workflow and compose.

3. Manager Dashboard (MAD)

Repository: forever-hotel/manager-dashboard (+ api-gateway 7c00d39edd, websocket-service f12ef41ba9, infra e3e17fcaac) | Branch: development | Tested revision: 8b2c63f459a93c217eb7cb502a2500e80181aa6e (baseline: bba65a0762286d4b0ac902865d7a2dd1e683e9c7) | CI date: 8 November 2026

Evidence: https://github.com/forever-hotel/manager-dashboard/actions/runs/42480400290, https://github.com/forever-hotel/manager-dashboard/actions/runs/42504148805, https://github.com/forever-hotel/api-gateway/actions/runs/42504067779, https://github.com/forever-hotel/websocket-service/actions/runs/42504249365

Register entries: 58; CI-suite-linked pass: 53; not tested/not verified: 5. Automated CI on 8b2c63f459 passes 534 backend unit tests, 5 changed-coverage tests, 203 PostgreSQL/RabbitMQ integration tests, 1 system integration test, 341 frontend Vitest tests, 6 frontend build/audit checks, 8 Playwright E2E tests, and all npm security audits.

TC-MAD-001 | PASS - CI suite

returns explicit check-in semantics, UTC boundaries and freshness

Requirement: MD-01 | Area: Booking totals | Level: Backend unit | Priority: P2

Preconditions	Test fixture and isolated CI dependencies defined in source file
Input / steps	Execute the named Jest/Vitest test with its checked-in fixtures
Expected result	Pass the exact named assertions shown in the linked test source
Recorded actual result	Matching unit, PostgreSQL E2E or Vitest suite passed in current CI; individual values are in source assertions
Test file / evidence	backend/src/analytics/booking-analytics.service.spec.ts Evidence reference https://github.com/forever-hotel/manager-dashboard/blob/bba65a0762286d4b0ac902865d7a2dd1e683e9c7/backend/src/analytics/booking-analytics.service.spec.ts

TC-MAD-002 | PASS - CI suite

TC-DDP-009-AC1: counts check-in dates with half-open day boundaries and hotel-local midnight

Requirement: MD-01 | Area: Booking totals | Level: Backend PostgreSQL E2E | Priority: P2

Preconditions	Test fixture and isolated CI dependencies defined in source file
Input / steps	Execute the named Jest/Vitest test with its checked-in fixtures
Expected result	Pass the exact named assertions shown in the linked test source
Recorded actual result	Matching unit, PostgreSQL E2E or Vitest suite passed in current CI; individual values are in source assertions
Test file / evidence	backend/test/bookings.e2e-spec.ts Evidence reference https://github.com/forever-hotel/manager-dashboard/blob/bba65a0762286d4b0ac902865d7a2dd1e683e9c7/backend/test/bookings.e2e-spec.ts

TC-MAD-003 | PASS - CI suite

TC-MAD-015-AC1: loads both metrics from one client and offers bar/line charts and a table

Requirement: MD-01 | Area: Booking dashboard | Level: Frontend component | Priority: P2

Preconditions	Test fixture and isolated CI dependencies defined in source file
Input / steps	Execute the named Jest/Vitest test with its checked-in fixtures
Expected result	Pass the exact named assertions shown in the linked test source
Recorded actual result	Matching unit, PostgreSQL E2E or Vitest suite passed in current CI; individual values are in source assertions
Test file / evidence	frontend/test/analytics.test.tsx Evidence reference https://github.com/forever-hotel/manager-dashboard/blob/bba65a0762286d4b0ac902865d7a2dd1e683e9c7/frontend/test/analytics.test.tsx

TC-MAD-004 | PASS - CI suite

TC-DDP-010-AC1: counts overlapping physical rooms once and excludes checkout day

Requirement: MD-02 | Area: Occupancy | Level: Backend PostgreSQL E2E | Priority: P2

Preconditions	Test fixture and isolated CI dependencies defined in source file
Input / steps	Execute the named Jest/Vitest test with its checked-in fixtures
Expected result	Pass the exact named assertions shown in the linked test source
Recorded actual result	Matching unit, PostgreSQL E2E or Vitest suite passed in current CI; individual values are in source assertions
Test file / evidence	backend/test/occupancy.e2e-spec.ts Evidence reference https://github.com/forever-hotel/manager-dashboard/blob/bba65a0762286d4b0ac902865d7a2dd1e683e9c7/backend/test/occupancy.e2e-spec.ts

TC-MAD-005 | PASS - CI suite

given zero capacity, returns a null rate and null source freshness

Requirement: MD-02 | Area: Occupancy | Level: Backend unit | Priority: P2

Preconditions	Test fixture and isolated CI dependencies defined in source file
Input / steps	Execute the named Jest/Vitest test with its checked-in fixtures
Expected result	Pass the exact named assertions shown in the linked test source
Recorded actual result	Matching unit, PostgreSQL E2E or Vitest suite passed in current CI; individual values are in source assertions
Test file / evidence	backend/src/analytics/occupancy-analytics.service.spec.ts Evidence reference https://github.com/forever-hotel/manager-dashboard/blob/bba65a0762286d4b0ac902865d7a2dd1e683e9c7/backend/src/analytics/occupancy-analytics.service.spec.ts

TC-MAD-006 | PASS - CI suite

TC-MAD-016-AC3: navigates leap February and does not display stale results

Requirement: MD-02 | Area: Calendar | Level: Frontend component | Priority: P2

Preconditions	Test fixture and isolated CI dependencies defined in source file
Input / steps	Execute the named Jest/Vitest test with its checked-in fixtures
Expected result	Pass the exact named assertions shown in the linked test source
Recorded actual result	Matching unit, PostgreSQL E2E or Vitest suite passed in current CI; individual values are in source assertions
Test file / evidence	frontend/test/occupancy.test.tsx Evidence reference https://github.com/forever-hotel/manager-dashboard/blob/bba65a0762286d4b0ac902865d7a2dd1e683e9c7/frontend/test/occupancy.test.tsx

TC-MAD-007 | PASS - CI suite

returns integer LKR cents and database-rounded percentage with shared timestamp

Requirement: MD-03 | Area: Revenue | Level: Backend unit | Priority: P2

Preconditions	Test fixture and isolated CI dependencies defined in source file
Input / steps	Execute the named Jest/Vitest test with its checked-in fixtures
Expected result	Pass the exact named assertions shown in the linked test source
Recorded actual result	Matching unit, PostgreSQL E2E or Vitest suite passed in current CI; individual values are in source assertions
Test file / evidence	backend/src/analytics/revenue-analytics.service.spec.ts Evidence reference https://github.com/forever-hotel/manager-dashboard/blob/bba65a0762286d4b0ac902865d7a2dd1e683e9c7/backend/src/analytics/revenue-analytics.service.spec.ts

TC-MAD-008 | PASS - CI suite

returns null and reason for zero previous revenue

Requirement: MD-03 | Area: Revenue | Level: Backend unit | Priority: P2

Preconditions	Test fixture and isolated CI dependencies defined in source file
Input / steps	Execute the named Jest/Vitest test with its checked-in fixtures
Expected result	Pass the exact named assertions shown in the linked test source
Recorded actual result	Matching unit, PostgreSQL E2E or Vitest suite passed in current CI; individual values are in source assertions
Test file / evidence	backend/src/analytics/revenue-analytics.service.spec.ts Evidence reference https://github.com/forever-hotel/manager-dashboard/blob/bba65a0762286d4b0ac902865d7a2dd1e683e9c7/backend/src/analytics/revenue-analytics.service.spec.ts

TC-MAD-009 | PASS - CI suite

AC1/AC2: includes only rates strictly below threshold and separates unknown rates

Requirement: MD-04 | Area: Low demand | Level: Backend unit | Priority: P2

Preconditions	Test fixture and isolated CI dependencies defined in source file
Input / steps	Execute the named Jest/Vitest test with its checked-in fixtures
Expected result	Pass the exact named assertions shown in the linked test source
Recorded actual result	Matching unit, PostgreSQL E2E or Vitest suite passed in current CI; individual values are in source assertions
Test file / evidence	backend/src/analytics/low-booking-alerts.service.spec.ts Evidence reference https://github.com/forever-hotel/manager-dashboard/blob/bba65a0762286d4b0ac902865d7a2dd1e683e9c7/backend/src/analytics/low-booking-alerts.service.spec.ts

TC-MAD-010 | PASS - CI suite

TC-MAD-057-AC3: distinguishes no alerts from unassessable dates

Requirement: MD-04 | Area: Low demand | Level: Frontend component | Priority: P2

Preconditions	Test fixture and isolated CI dependencies defined in source file
Input / steps	Execute the named Jest/Vitest test with its checked-in fixtures
Expected result	Pass the exact named assertions shown in the linked test source
Recorded actual result	Matching unit, PostgreSQL E2E or Vitest suite passed in current CI; individual values are in source assertions
Test file / evidence	frontend/test/low-demand.test.tsx Evidence reference https://github.com/forever-hotel/manager-dashboard/blob/bba65a0762286d4b0ac902865d7a2dd1e683e9c7/frontend/test/low-demand.test.tsx

TC-MAD-011 | PASS - CI suite

normalizes only ASCII codes and validates positive integer currency/limits

Requirement: MD-05 | Area: Create promotions | Level: Backend unit | Priority: P2

Preconditions	Test fixture and isolated CI dependencies defined in source file
Input / steps	Execute the named Jest/Vitest test with its checked-in fixtures
Expected result	Pass the exact named assertions shown in the linked test source
Recorded actual result	Matching unit, PostgreSQL E2E or Vitest suite passed in current CI; individual values are in source assertions
Test file / evidence	backend/src/promotions/promotions.spec.ts Evidence reference https://github.com/forever-hotel/manager-dashboard/blob/bba65a0762286d4b0ac902865d7a2dd1e683e9c7/backend/src/promotions/promotions.spec.ts

TC-MAD-012 | PASS - CI suite

rejects unknown room types and legacy duplicates before insert

Requirement: MD-05 | Area: Create promotions | Level: Backend unit | Priority: P2

Preconditions	Test fixture and isolated CI dependencies defined in source file
Input / steps	Execute the named Jest/Vitest test with its checked-in fixtures
Expected result	Pass the exact named assertions shown in the linked test source
Recorded actual result	Matching unit, PostgreSQL E2E or Vitest suite passed in current CI; individual values are in source assertions
Test file / evidence	backend/src/promotions/promotions.spec.ts Evidence reference https://github.com/forever-hotel/manager-dashboard/blob/bba65a0762286d4b0ac902865d7a2dd1e683e9c7/backend/src/promotions/promotions.spec.ts

TC-MAD-013 | PASS - CI suite

TC-MAD-020-AC1: saves a draft inactive atomically and exposes publication failure safely

Requirement: MD-05 | Area: Create promotions | Level: Backend PostgreSQL E2E | Priority: P2

Preconditions	Test fixture and isolated CI dependencies defined in source file
Input / steps	Execute the named Jest/Vitest test with its checked-in fixtures
Expected result	Pass the exact named assertions shown in the linked test source
Recorded actual result	Matching unit, PostgreSQL E2E or Vitest suite passed in current CI; individual values are in source assertions
Test file / evidence	backend/test/promotions.e2e-spec.ts Evidence reference https://github.com/forever-hotel/manager-dashboard/blob/bba65a0762286d4b0ac902865d7a2dd1e683e9c7/backend/test/promotions.e2e-spec.ts

TC-MAD-014 | PASS - CI suite

replaces editable fields and room associations, retaining redemptions

Requirement: MD-05b | Area: Promotion management | Level: Backend unit | Priority: P2

Preconditions	Test fixture and isolated CI dependencies defined in source file
Input / steps	Execute the named Jest/Vitest test with its checked-in fixtures
Expected result	Pass the exact named assertions shown in the linked test source
Recorded actual result	Matching unit, PostgreSQL E2E or Vitest suite passed in current CI; individual values are in source assertions
Test file / evidence	backend/src/promotions/promotions.spec.ts Evidence reference https://github.com/forever-hotel/manager-dashboard/blob/bba65a0762286d4b0ac902865d7a2dd1e683e9c7/backend/src/promotions/promotions.spec.ts

TC-MAD-015 | PASS - CI suite

TC-MAD-020-AC2: reloads a conflicting version and preserves unsaved input

Requirement: MD-05b | Area: Promotion management | Level: Frontend component | Priority: P2

Preconditions	Test fixture and isolated CI dependencies defined in source file
Input / steps	Execute the named Jest/Vitest test with its checked-in fixtures
Expected result	Pass the exact named assertions shown in the linked test source
Recorded actual result	Matching unit, PostgreSQL E2E or Vitest suite passed in current CI; individual values are in source assertions
Test file / evidence	frontend/test/promotions.test.tsx Evidence reference https://github.com/forever-hotel/manager-dashboard/blob/bba65a0762286d4b0ac902865d7a2dd1e683e9c7/frontend/test/promotions.test.tsx

TC-MAD-016 | PASS - CI suite

TC-MAD-040-AC1: routes a committed publication through a real confirmed broker contract queue

Requirement: MD-05c | Area: Promotion publication | Level: Backend PostgreSQL E2E | Priority: P1

Preconditions	Test fixture and isolated CI dependencies defined in source file
Input / steps	Execute the named Jest/Vitest test with its checked-in fixtures
Expected result	Pass the exact named assertions shown in the linked test source
Recorded actual result	Matching unit, PostgreSQL E2E or Vitest suite passed in current CI; individual values are in source assertions
Test file / evidence	backend/test/promotions.e2e-spec.ts Evidence reference https://github.com/forever-hotel/manager-dashboard/blob/bba65a0762286d4b0ac902865d7a2dd1e683e9c7/backend/test/promotions.e2e-spec.ts

TC-MAD-017 | PASS - CI suite

retains failed work with capped exponential retry and a fixed safe error

Requirement: MD-05c | Area: Promotion publication | Level: Backend unit | Priority: P1

Preconditions	Test fixture and isolated CI dependencies defined in source file
Input / steps	Execute the named Jest/Vitest test with its checked-in fixtures
Expected result	Pass the exact named assertions shown in the linked test source
Recorded actual result	Matching unit, PostgreSQL E2E or Vitest suite passed in current CI; individual values are in source assertions
Test file / evidence	backend/src/promotions/promotion-publication.spec.ts Evidence reference https://github.com/forever-hotel/manager-dashboard/blob/bba65a0762286d4b0ac902865d7a2dd1e683e9c7/backend/src/promotions/promotion-publication.spec.ts

TC-MAD-018 | PASS - CI suite

TC-MAD-023-AC1: counts each order once, historical quantities and half-hour-offset peak hours

Requirement: MD-06 | Area: Food analytics | Level: Backend PostgreSQL E2E | Priority: P2

Preconditions	Test fixture and isolated CI dependencies defined in source file
Input / steps	Execute the named Jest/Vitest test with its checked-in fixtures
Expected result	Pass the exact named assertions shown in the linked test source
Recorded actual result	Matching unit, PostgreSQL E2E or Vitest suite passed in current CI; individual values are in source assertions
Test file / evidence	backend/test/food-analytics.e2e-spec.ts Evidence reference https://github.com/forever-hotel/manager-dashboard/blob/bba65a0762286d4b0ac902865d7a2dd1e683e9c7/backend/test/food-analytics.e2e-spec.ts

TC-MAD-019 | PASS - CI suite

returns zero-filled empty results and the documented policy

Requirement: MD-06 | Area: Food analytics | Level: Backend unit | Priority: P2

Preconditions	Test fixture and isolated CI dependencies defined in source file
Input / steps	Execute the named Jest/Vitest test with its checked-in fixtures
Expected result	Pass the exact named assertions shown in the linked test source
Recorded actual result	Matching unit, PostgreSQL E2E or Vitest suite passed in current CI; individual values are in source assertions
Test file / evidence	backend/src/analytics/food-analytics.service.spec.ts Evidence reference https://github.com/forever-hotel/manager-dashboard/blob/bba65a0762286d4b0ac902865d7a2dd1e683e9c7/backend/src/analytics/food-analytics.service.spec.ts

TC-MAD-020 | PASS - CI suite

TC-MAD-041-AC1: maps four states, independent item counts, stable pagination and delivered retention

Requirement: MD-07 | Area: Kitchen statuses | Level: Backend PostgreSQL E2E | Priority: P2

Preconditions	Test fixture and isolated CI dependencies defined in source file
Input / steps	Execute the named Jest/Vitest test with its checked-in fixtures
Expected result	Pass the exact named assertions shown in the linked test source
Recorded actual result	Matching unit, PostgreSQL E2E or Vitest suite passed in current CI; individual values are in source assertions
Test file / evidence	backend/test/kitchen-realtime.e2e-spec.ts Evidence reference https://github.com/forever-hotel/manager-dashboard/blob/bba65a0762286d4b0ac902865d7a2dd1e683e9c7/backend/test/kitchen-realtime.e2e-spec.ts

TC-MAD-021 | PASS - CI suite

TC-MAD-026-AC2: reconciles a status change exactly once and ignores duplicate/older versions

Requirement: MD-07 | Area: Kitchen statuses | Level: Frontend component | Priority: P2

Preconditions	Test fixture and isolated CI dependencies defined in source file
Input / steps	Execute the named Jest/Vitest test with its checked-in fixtures
Expected result	Pass the exact named assertions shown in the linked test source
Recorded actual result	Matching unit, PostgreSQL E2E or Vitest suite passed in current CI; individual values are in source assertions
Test file / evidence	frontend/test/kitchen.test.tsx Evidence reference https://github.com/forever-hotel/manager-dashboard/blob/bba65a0762286d4b0ac902865d7a2dd1e683e9c7/frontend/test/kitchen.test.tsx

TC-MAD-022 | PASS - CI suite

returns explicit cohorts, null average and zero-filled states

Requirement: MD-08 | Area: Service requests | Level: Backend unit | Priority: P2

Preconditions	Test fixture and isolated CI dependencies defined in source file
Input / steps	Execute the named Jest/Vitest test with its checked-in fixtures
Expected result	Pass the exact named assertions shown in the linked test source
Recorded actual result	Matching unit, PostgreSQL E2E or Vitest suite passed in current CI; individual values are in source assertions
Test file / evidence	backend/src/analytics/service-request-analytics.service.spec.ts Evidence reference https://github.com/forever-hotel/manager-dashboard/blob/bba65a0762286d4b0ac902865d7a2dd1e683e9c7/backend/src/analytics/service-request-analytics.service.spec.ts

TC-MAD-023 | PASS - CI suite

sanitizes missing or failed source results and forwards controller periods

Requirement: MD-08 | Area: Service requests | Level: Backend unit | Priority: P2

Preconditions	Test fixture and isolated CI dependencies defined in source file
Input / steps	Execute the named Jest/Vitest test with its checked-in fixtures
Expected result	Pass the exact named assertions shown in the linked test source
Recorded actual result	Matching unit, PostgreSQL E2E or Vitest suite passed in current CI; individual values are in source assertions
Test file / evidence	backend/src/analytics/service-request-analytics.service.spec.ts Evidence reference https://github.com/forever-hotel/manager-dashboard/blob/bba65a0762286d4b0ac902865d7a2dd1e683e9c7/backend/src/analytics/service-request-analytics.service.spec.ts

TC-MAD-024 | PASS - CI suite

TC-MAD-021-AC1: lists safe filtered/paginated complaints including a null worker

Requirement: MD-09 | Area: Complaint register | Level: Backend PostgreSQL E2E | Priority: P2

Preconditions	Test fixture and isolated CI dependencies defined in source file
Input / steps	Execute the named Jest/Vitest test with its checked-in fixtures
Expected result	Pass the exact named assertions shown in the linked test source
Recorded actual result	Matching unit, PostgreSQL E2E or Vitest suite passed in current CI; individual values are in source assertions
Test file / evidence	backend/test/complaints.e2e-spec.ts Evidence reference https://github.com/forever-hotel/manager-dashboard/blob/bba65a0762286d4b0ac902865d7a2dd1e683e9c7/backend/test/complaints.e2e-spec.ts

TC-MAD-025 | PASS - CI suite

TC-MAD-022-AC1: displays required fields, null worker and exact filters

Requirement: MD-09 | Area: Complaint register | Level: Frontend component | Priority: P2

Preconditions	Test fixture and isolated CI dependencies defined in source file
Input / steps	Execute the named Jest/Vitest test with its checked-in fixtures
Expected result	Pass the exact named assertions shown in the linked test source
Recorded actual result	Matching unit, PostgreSQL E2E or Vitest suite passed in current CI; individual values are in source assertions
Test file / evidence	frontend/test/complaints.test.tsx Evidence reference https://github.com/forever-hotel/manager-dashboard/blob/bba65a0762286d4b0ac902865d7a2dd1e683e9c7/frontend/test/complaints.test.tsx

TC-MAD-026 | PASS - CI suite

TC-MAD-021-AC2: records actor/time, both transitions, private notes and note-free events

Requirement: MD-09b | Area: Complaint resolution | Level: Backend PostgreSQL E2E | Priority: P1

Preconditions	Test fixture and isolated CI dependencies defined in source file
Input / steps	Execute the named Jest/Vitest test with its checked-in fixtures
Expected result	Pass the exact named assertions shown in the linked test source
Recorded actual result	Matching unit, PostgreSQL E2E or Vitest suite passed in current CI; individual values are in source assertions
Test file / evidence	backend/test/complaints.e2e-spec.ts Evidence reference https://github.com/forever-hotel/manager-dashboard/blob/bba65a0762286d4b0ac902865d7a2dd1e683e9c7/backend/test/complaints.e2e-spec.ts

TC-MAD-027 | PASS - CI suite

allows exactly two forward transitions

Requirement: MD-09b | Area: Complaint resolution | Level: Backend unit | Priority: P1

Preconditions	Test fixture and isolated CI dependencies defined in source file
Input / steps	Execute the named Jest/Vitest test with its checked-in fixtures
Expected result	Pass the exact named assertions shown in the linked test source
Recorded actual result	Matching unit, PostgreSQL E2E or Vitest suite passed in current CI; individual values are in source assertions
Test file / evidence	backend/src/complaints/complaints.spec.ts Evidence reference https://github.com/forever-hotel/manager-dashboard/blob/bba65a0762286d4b0ac902865d7a2dd1e683e9c7/backend/src/complaints/complaints.spec.ts

TC-MAD-028 | PASS - CI suite

TC-MAD-025-AC1/AC3: independently aggregates historical completions and current referenced complaints

Requirement: MD-10 | Area: Worker metrics | Level: Backend PostgreSQL E2E | Priority: P2

Preconditions	Test fixture and isolated CI dependencies defined in source file
Input / steps	Execute the named Jest/Vitest test with its checked-in fixtures
Expected result	Pass the exact named assertions shown in the linked test source
Recorded actual result	Matching unit, PostgreSQL E2E or Vitest suite passed in current CI; individual values are in source assertions
Test file / evidence	backend/test/worker-performance.e2e-spec.ts Evidence reference https://github.com/forever-hotel/manager-dashboard/blob/bba65a0762286d4b0ac902865d7a2dd1e683e9c7/backend/test/worker-performance.e2e-spec.ts

TC-MAD-029 | PASS - CI suite

Given a worker without activity, returns zero counts and a null average with explicit cohorts

Requirement: MD-10 | Area: Worker metrics | Level: Backend unit | Priority: P2

Preconditions	Test fixture and isolated CI dependencies defined in source file
Input / steps	Execute the named Jest/Vitest test with its checked-in fixtures
Expected result	Pass the exact named assertions shown in the linked test source
Recorded actual result	Matching unit, PostgreSQL E2E or Vitest suite passed in current CI; individual values are in source assertions
Test file / evidence	backend/src/analytics/worker-performance.service.spec.ts Evidence reference https://github.com/forever-hotel/manager-dashboard/blob/bba65a0762286d4b0ac902865d7a2dd1e683e9c7/backend/src/analytics/worker-performance.service.spec.ts

TC-MAD-030 | PASS - CI suite

TC-MAD-030-AC3: supports empty reports and real PDF attachments for every type

Requirement: MD-11 | Area: Report export | Level: Backend PostgreSQL E2E | Priority: P2

Preconditions	Test fixture and isolated CI dependencies defined in source file
Input / steps	Execute the named Jest/Vitest test with its checked-in fixtures
Expected result	Pass the exact named assertions shown in the linked test source
Recorded actual result	Matching unit, PostgreSQL E2E or Vitest suite passed in current CI; individual values are in source assertions
Test file / evidence	backend/test/reports.e2e-spec.ts Evidence reference https://github.com/forever-hotel/manager-dashboard/blob/bba65a0762286d4b0ac902865d7a2dd1e683e9c7/backend/test/reports.e2e-spec.ts

TC-MAD-031 | PASS - CI suite

quotes commas/quotes/newlines and neutralizes formula prefixes after whitespace

Requirement: MD-11 | Area: Report export | Level: Backend unit | Priority: P2

Preconditions	Test fixture and isolated CI dependencies defined in source file
Input / steps	Execute the named Jest/Vitest test with its checked-in fixtures
Expected result	Pass the exact named assertions shown in the linked test source
Recorded actual result	Matching unit, PostgreSQL E2E or Vitest suite passed in current CI; individual values are in source assertions
Test file / evidence	backend/src/reports/reports.spec.ts Evidence reference https://github.com/forever-hotel/manager-dashboard/blob/bba65a0762286d4b0ac902865d7a2dd1e683e9c7/backend/src/reports/reports.spec.ts

TC-MAD-032 | PASS - CI suite

AC1: sends current filters, names the file and revokes temporary URLs

Requirement: MD-11 | Area: Report export | Level: Frontend component | Priority: P2

Preconditions	Test fixture and isolated CI dependencies defined in source file
Input / steps	Execute the named Jest/Vitest test with its checked-in fixtures
Expected result	Pass the exact named assertions shown in the linked test source
Recorded actual result	Matching unit, PostgreSQL E2E or Vitest suite passed in current CI; individual values are in source assertions
Test file / evidence	frontend/test/reports.test.tsx Evidence reference https://github.com/forever-hotel/manager-dashboard/blob/bba65a0762286d4b0ac902865d7a2dd1e683e9c7/frontend/test/reports.test.tsx

TC-MAD-033 | PASS - CI suite

TC-MAD-027-AC1: exposes every room with its stable ID, type and all four mapped states

Requirement: MD-12 | Area: Room status | Level: Backend PostgreSQL E2E | Priority: P2

Preconditions	Test fixture and isolated CI dependencies defined in source file
Input / steps	Execute the named Jest/Vitest test with its checked-in fixtures
Expected result	Pass the exact named assertions shown in the linked test source
Recorded actual result	Matching unit, PostgreSQL E2E or Vitest suite passed in current CI; individual values are in source assertions
Test file / evidence	backend/test/rooms.e2e-spec.ts Evidence reference https://github.com/forever-hotel/manager-dashboard/blob/bba65a0762286d4b0ac902865d7a2dd1e683e9c7/backend/test/rooms.e2e-spec.ts

TC-MAD-034 | PASS - CI suite

TC-MAD-027-AC2: publishes a committed authoritative room update through RabbitMQ to an authorized socket

Requirement: MD-12 | Area: Room status | Level: Backend PostgreSQL E2E | Priority: P2

Preconditions	Test fixture and isolated CI dependencies defined in source file
Input / steps	Execute the named Jest/Vitest test with its checked-in fixtures
Expected result	Pass the exact named assertions shown in the linked test source
Recorded actual result	Matching unit, PostgreSQL E2E or Vitest suite passed in current CI; individual values are in source assertions
Test file / evidence	backend/test/rooms.e2e-spec.ts Evidence reference https://github.com/forever-hotel/manager-dashboard/blob/bba65a0762286d4b0ac902865d7a2dd1e683e9c7/backend/test/rooms.e2e-spec.ts

TC-MAD-035 | PASS - CI suite

AC2: rereads authoritative data on events and reconnect

Requirement: MD-12 | Area: Room status | Level: Frontend component | Priority: P2

Preconditions	Test fixture and isolated CI dependencies defined in source file
Input / steps	Execute the named Jest/Vitest test with its checked-in fixtures
Expected result	Pass the exact named assertions shown in the linked test source
Recorded actual result	Matching unit, PostgreSQL E2E or Vitest suite passed in current CI; individual values are in source assertions
Test file / evidence	frontend/test/rooms.test.tsx Evidence reference https://github.com/forever-hotel/manager-dashboard/blob/bba65a0762286d4b0ac902865d7a2dd1e683e9c7/frontend/test/rooms.test.tsx

TC-MAD-036 | PASS - CI suite

TC-MAD-034-AC1/AC2: persists once, coalesces versions and clears on recovery without source writes

Requirement: MD-13 | Area: Notifications | Level: Backend PostgreSQL E2E | Priority: P2

Preconditions	Test fixture and isolated CI dependencies defined in source file
Input / steps	Execute the named Jest/Vitest test with its checked-in fixtures
Expected result	Pass the exact named assertions shown in the linked test source
Recorded actual result	Matching unit, PostgreSQL E2E or Vitest suite passed in current CI; individual values are in source assertions
Test file / evidence	backend/test/stock.e2e-spec.ts Evidence reference https://github.com/forever-hotel/manager-dashboard/blob/bba65a0762286d4b0ac902865d7a2dd1e683e9c7/backend/test/stock.e2e-spec.ts

TC-MAD-037 | PASS - CI suite

projects safe deduplicated issue links, consistent counts and empty/beyond-last pages

Requirement: MD-13 | Area: Notifications | Level: Backend unit | Priority: P2

Preconditions	Test fixture and isolated CI dependencies defined in source file
Input / steps	Execute the named Jest/Vitest test with its checked-in fixtures
Expected result	Pass the exact named assertions shown in the linked test source
Recorded actual result	Matching unit, PostgreSQL E2E or Vitest suite passed in current CI; individual values are in source assertions
Test file / evidence	backend/src/notifications/notifications.spec.ts Evidence reference https://github.com/forever-hotel/manager-dashboard/blob/bba65a0762286d4b0ac902865d7a2dd1e683e9c7/backend/src/notifications/notifications.spec.ts

TC-MAD-038 | PASS - CI suite

stores a cost-12 hash and encrypted delivery, returning only identifiers and state

Requirement: MD-14 | Area: Receptionist provisioning | Level: Backend unit | Priority: P1

Preconditions	Test fixture and isolated CI dependencies defined in source file
Input / steps	Execute the named Jest/Vitest test with its checked-in fixtures
Expected result	Pass the exact named assertions shown in the linked test source
Recorded actual result	Matching unit, PostgreSQL E2E or Vitest suite passed in current CI; individual values are in source assertions
Test file / evidence	backend/src/staff/staff.spec.ts Evidence reference https://github.com/forever-hotel/manager-dashboard/blob/bba65a0762286d4b0ac902865d7a2dd1e683e9c7/backend/src/staff/staff.spec.ts

TC-MAD-039 | PASS - CI suite

TC-MAD-012-AC3: concurrent requests enforce email and username uniqueness

Requirement: MD-14 | Area: Receptionist provisioning | Level: Backend PostgreSQL E2E | Priority: P1

Preconditions	Test fixture and isolated CI dependencies defined in source file
Input / steps	Execute the named Jest/Vitest test with its checked-in fixtures
Expected result	Pass the exact named assertions shown in the linked test source
Recorded actual result	Matching unit, PostgreSQL E2E or Vitest suite passed in current CI; individual values are in source assertions
Test file / evidence	backend/test/staff.e2e-spec.ts Evidence reference https://github.com/forever-hotel/manager-dashboard/blob/bba65a0762286d4b0ac902865d7a2dd1e683e9c7/backend/test/staff.e2e-spec.ts

TC-MAD-040 | PASS - CI suite

authenticates ciphertext and delivery identity without storing plaintext

Requirement: MD-15 | Area: Worker provisioning | Level: Backend unit | Priority: P1

Preconditions	Test fixture and isolated CI dependencies defined in source file
Input / steps	Execute the named Jest/Vitest test with its checked-in fixtures
Expected result	Pass the exact named assertions shown in the linked test source
Recorded actual result	Matching unit, PostgreSQL E2E or Vitest suite passed in current CI; individual values are in source assertions
Test file / evidence	backend/src/staff/staff.spec.ts Evidence reference https://github.com/forever-hotel/manager-dashboard/blob/bba65a0762286d4b0ac902865d7a2dd1e683e9c7/backend/src/staff/staff.spec.ts

TC-MAD-041 | PASS - CI suite

TC-MAD-012-AC2: duplicate email returns 409 without another delivery

Requirement: MD-15 | Area: Worker provisioning | Level: Backend PostgreSQL E2E | Priority: P1

Preconditions	Test fixture and isolated CI dependencies defined in source file
Input / steps	Execute the named Jest/Vitest test with its checked-in fixtures
Expected result	Pass the exact named assertions shown in the linked test source
Recorded actual result	Matching unit, PostgreSQL E2E or Vitest suite passed in current CI; individual values are in source assertions
Test file / evidence	backend/test/staff.e2e-spec.ts Evidence reference https://github.com/forever-hotel/manager-dashboard/blob/bba65a0762286d4b0ac902865d7a2dd1e683e9c7/backend/test/staff.e2e-spec.ts

TC-MAD-042 | PASS - CI suite

AC1: supports all roles and required fields and displays queued delivery

Requirement: MD-15b | Area: Kitchen staff provisioning | Level: Frontend component | Priority: P2

Preconditions	Test fixture and isolated CI dependencies defined in source file
Input / steps	Execute the named Jest/Vitest test with its checked-in fixtures
Expected result	Pass the exact named assertions shown in the linked test source
Recorded actual result	Matching unit, PostgreSQL E2E or Vitest suite passed in current CI; individual values are in source assertions
Test file / evidence	frontend/test/staff.test.tsx Evidence reference https://github.com/forever-hotel/manager-dashboard/blob/bba65a0762286d4b0ac902865d7a2dd1e683e9c7/frontend/test/staff.test.tsx

TC-MAD-043 | PASS - CI suite

validates optional and enabled email configuration without exposing values

Requirement: MD-15b | Area: Kitchen staff provisioning | Level: Backend unit | Priority: P2

Preconditions	Test fixture and isolated CI dependencies defined in source file
Input / steps	Execute the named Jest/Vitest test with its checked-in fixtures
Expected result	Pass the exact named assertions shown in the linked test source
Recorded actual result	Matching unit, PostgreSQL E2E or Vitest suite passed in current CI; individual values are in source assertions
Test file / evidence	backend/src/staff/staff.spec.ts Evidence reference https://github.com/forever-hotel/manager-dashboard/blob/bba65a0762286d4b0ac902865d7a2dd1e683e9c7/backend/src/staff/staff.spec.ts

TC-MAD-044 | PASS - CI suite

given active staff, replaces credentials, increments version and records a secret-free audit

Requirement: MD-16 | Area: Password reset | Level: Backend unit | Priority: P1

Preconditions	Test fixture and isolated CI dependencies defined in source file
Input / steps	Execute the named Jest/Vitest test with its checked-in fixtures
Expected result	Pass the exact named assertions shown in the linked test source
Recorded actual result	Matching unit, PostgreSQL E2E or Vitest suite passed in current CI; individual values are in source assertions
Test file / evidence	backend/src/staff/staff-reset.spec.ts Evidence reference https://github.com/forever-hotel/manager-dashboard/blob/bba65a0762286d4b0ac902865d7a2dd1e683e9c7/backend/src/staff/staff-reset.spec.ts

TC-MAD-045 | PASS - CI suite

given no encryption key, fails before updating the account

Requirement: MD-16 | Area: Password reset | Level: Backend unit | Priority: P1

Preconditions	Test fixture and isolated CI dependencies defined in source file
Input / steps	Execute the named Jest/Vitest test with its checked-in fixtures
Expected result	Pass the exact named assertions shown in the linked test source
Recorded actual result	Matching unit, PostgreSQL E2E or Vitest suite passed in current CI; individual values are in source assertions
Test file / evidence	backend/src/staff/staff-reset.spec.ts Evidence reference https://github.com/forever-hotel/manager-dashboard/blob/bba65a0762286d4b0ac902865d7a2dd1e683e9c7/backend/src/staff/staff-reset.spec.ts

TC-MAD-046 | PASS - CI suite

accepts partial normalized profile changes and deactivation

Requirement: MD-16b | Area: Staff management | Level: Backend unit | Priority: P1

Preconditions	Test fixture and isolated CI dependencies defined in source file
Input / steps	Execute the named Jest/Vitest test with its checked-in fixtures
Expected result	Pass the exact named assertions shown in the linked test source
Recorded actual result	Matching unit, PostgreSQL E2E or Vitest suite passed in current CI; individual values are in source assertions
Test file / evidence	backend/src/staff/staff-profiles.spec.ts Evidence reference https://github.com/forever-hotel/manager-dashboard/blob/bba65a0762286d4b0ac902865d7a2dd1e683e9c7/backend/src/staff/staff-profiles.spec.ts

TC-MAD-047 | PASS - CI suite

AC2: edits fields and roles with the current version then refreshes

Requirement: MD-16b | Area: Staff management | Level: Frontend component | Priority: P1

Preconditions	Test fixture and isolated CI dependencies defined in source file
Input / steps	Execute the named Jest/Vitest test with its checked-in fixtures
Expected result	Pass the exact named assertions shown in the linked test source
Recorded actual result	Matching unit, PostgreSQL E2E or Vitest suite passed in current CI; individual values are in source assertions
Test file / evidence	frontend/test/staff.test.tsx Evidence reference https://github.com/forever-hotel/manager-dashboard/blob/bba65a0762286d4b0ac902865d7a2dd1e683e9c7/frontend/test/staff.test.tsx

TC-MAD-048 | PASS - CI suite

TC-MAD-042-AC1: filters and paginates stable tasks with worker and completion details

Requirement: MD-17 | Area: Read-only task queue | Level: Backend PostgreSQL E2E | Priority: P2

Preconditions	Test fixture and isolated CI dependencies defined in source file
Input / steps	Execute the named Jest/Vitest test with its checked-in fixtures
Expected result	Pass the exact named assertions shown in the linked test source
Recorded actual result	Matching unit, PostgreSQL E2E or Vitest suite passed in current CI; individual values are in source assertions
Test file / evidence	backend/test/tasks.e2e-spec.ts Evidence reference https://github.com/forever-hotel/manager-dashboard/blob/bba65a0762286d4b0ac902865d7a2dd1e683e9c7/backend/test/tasks.e2e-spec.ts

TC-MAD-049 | PASS - CI suite

AC1/AC3: exposes all task groups and worker details without mutation controls

Requirement: MD-17 | Area: Read-only task queue | Level: Frontend component | Priority: P2

Preconditions	Test fixture and isolated CI dependencies defined in source file
Input / steps	Execute the named Jest/Vitest test with its checked-in fixtures
Expected result	Pass the exact named assertions shown in the linked test source
Recorded actual result	Matching unit, PostgreSQL E2E or Vitest suite passed in current CI; individual values are in source assertions
Test file / evidence	frontend/test/tasks.test.tsx Evidence reference https://github.com/forever-hotel/manager-dashboard/blob/bba65a0762286d4b0ac902865d7a2dd1e683e9c7/frontend/test/tasks.test.tsx

TC-MAD-050 | PASS - CI suite

persists failed-login counters and locks after five guesses

Requirement: NFR-12 | Area: Authentication | Level: Backend unit | Priority: P1

Preconditions	Test fixture and isolated CI dependencies defined in source file
Input / steps	Execute the named Jest/Vitest test with its checked-in fixtures
Expected result	Pass the exact named assertions shown in the linked test source
Recorded actual result	Matching unit, PostgreSQL E2E or Vitest suite passed in current CI; individual values are in source assertions
Test file / evidence	backend/src/auth/auth.service.spec.ts Evidence reference https://github.com/forever-hotel/manager-dashboard/blob/bba65a0762286d4b0ac902865d7a2dd1e683e9c7/backend/src/auth/auth.service.spec.ts

TC-MAD-051 | PASS - CI suite

redirects direct links without a session and preserves the destination

Requirement: NFR-12 | Area: Access control | Level: Frontend component | Priority: P1

Preconditions	Test fixture and isolated CI dependencies defined in source file
Input / steps	Execute the named Jest/Vitest test with its checked-in fixtures
Expected result	Pass the exact named assertions shown in the linked test source
Recorded actual result	Matching unit, PostgreSQL E2E or Vitest suite passed in current CI; individual values are in source assertions
Test file / evidence	frontend/test/protection.test.ts Evidence reference https://github.com/forever-hotel/manager-dashboard/blob/bba65a0762286d4b0ac902865d7a2dd1e683e9c7/frontend/test/protection.test.ts

TC-MAD-052 | PASS - CI suite

protects pages and BFF routes without changing presentation or blocking sockets

Requirement: NFR-12 | Area: Secure headers | Level: Frontend component | Priority: P1

Preconditions	Test fixture and isolated CI dependencies defined in source file
Input / steps	Execute the named Jest/Vitest test with its checked-in fixtures
Expected result	Pass the exact named assertions shown in the linked test source
Recorded actual result	Matching unit, PostgreSQL E2E or Vitest suite passed in current CI; individual values are in source assertions
Test file / evidence	frontend/test/security-headers.test.ts Evidence reference https://github.com/forever-hotel/manager-dashboard/blob/bba65a0762286d4b0ac902865d7a2dd1e683e9c7/frontend/test/security-headers.test.ts

TC-MAD-053 | PASS - CI suite

TC-MAD-041-AC1: maps four states, independent item counts, stable pagination and delivered retention

Requirement: MD-07 | Area: Realtime state mapping (not latency testing) | Level: Backend PostgreSQL E2E | Priority: P2

Preconditions	Test fixture and isolated CI dependencies defined in source file
Input / steps	Execute the named Jest/Vitest test with its checked-in fixtures
Expected result	Pass the exact named assertions shown in the linked test source
Recorded actual result	Matching unit, PostgreSQL E2E or Vitest suite passed in current CI; individual values are in source assertions
Test file / evidence	backend/test/kitchen-realtime.e2e-spec.ts Evidence reference https://github.com/forever-hotel/manager-dashboard/blob/bba65a0762286d4b0ac902865d7a2dd1e683e9c7/backend/test/kitchen-realtime.e2e-spec.ts

TC-MAD-054 | NOT VERIFIED

Live publication/redeem on deployed Hotel Website

Requirement: MD-05c | Area: Integration acceptance | Level: Cross-service acceptance | Priority: P1

Preconditions	Integrated deployment with approved test identities and separate test data
Input / steps	Execute scenario in a controlled integrated staging environment
Expected result	End-to-end business flow meets SRS acceptance requirement
Recorded actual result	No public cross-service acceptance record inspected.
Test file / evidence	Not independently demonstrated in retrieved CI evidence Evidence reference https://github.com/forever-hotel/manager-dashboard/actions

TC-MAD-055 | NOT VERIFIED

Live KMS event flow after independent service deployments

Requirement: MD-07 | Area: Integration acceptance | Level: Cross-service acceptance | Priority: P1

Preconditions	Integrated deployment with approved test identities and separate test data
Input / steps	Execute scenario in a controlled integrated staging environment
Expected result	End-to-end business flow meets SRS acceptance requirement
Recorded actual result	Isolated CI RabbitMQ tests do not establish fully deployed six-subsystem availability.
Test file / evidence	Not independently demonstrated in retrieved CI evidence Evidence reference https://github.com/forever-hotel/manager-dashboard/actions

TC-MAD-056 | NOT VERIFIED

Production email delivery to provisioned staff

Requirement: MD-14/15/15b | Area: Integration acceptance | Level: External-provider acceptance | Priority: P1

Preconditions	Integrated deployment with approved test identities and separate test data
Input / steps	Execute scenario in a controlled integrated staging environment
Expected result	End-to-end business flow meets SRS acceptance requirement
Recorded actual result	CI tests use controlled/fixture delivery; provider acceptance not independently confirmed.
Test file / evidence	Not independently demonstrated in retrieved CI evidence Evidence reference https://github.com/forever-hotel/manager-dashboard/actions

TC-MAD-057 | NOT VERIFIED

Concurrent real receptionist/worker/manager sessions

Requirement: MD-12/17 | Area: Integration acceptance | Level: Integrated staging acceptance | Priority: P1

Preconditions	Integrated deployment with approved test identities and separate test data
Input / steps	Execute scenario in a controlled integrated staging environment
Expected result	End-to-end business flow meets SRS acceptance requirement
Recorded actual result	No separately approved complete hotel staging UAT evidence inspected.
Test file / evidence	Not independently demonstrated in retrieved CI evidence Evidence reference https://github.com/forever-hotel/manager-dashboard/actions

TC-MAD-058 | NOT VERIFIED

Verified 100/300-user load and end-to-end UI timing

Requirement: NFR-01/02/03/14 | Area: Integration acceptance | Level: Performance acceptance | Priority: P1

Preconditions	Integrated deployment with approved test identities and separate test data
Input / steps	Execute scenario in a controlled integrated staging environment
Expected result	End-to-end business flow meets SRS acceptance requirement
Recorded actual result	Tools exist but signed, correlated measurements were not independently retrieved.
Test file / evidence	Not independently demonstrated in retrieved CI evidence Evidence reference https://github.com/forever-hotel/manager-dashboard/actions

4. Front Desk System (FDS)

Repository: forever-hotel/front-desk | Branch: develop (QA branch: feature/84-fds-final-qa-evidence) | Tested revision: 4b5ef57d4c4ce03c9ab2fe78f42b94eff96838cd (QA snapshot: e590326 / 1065617 / a6050dd) | CI date: 8 November 2026

Evidence: https://github.com/forever-hotel/front-desk/actions/runs/42506217082, https://github.com/forever-hotel/front-desk/actions/runs/42506217056, https://github.com/forever-hotel/front-desk/actions/runs/40530366600, https://github.com/forever-hotel/front-desk/actions/runs/40534533200

Register entries: 43; CI-suite-linked pass: 34; not tested/not verified: 9 (automated develop CI passes 949 tests: 531 BE unit, 56 DB integration, 143 HTTP E2E, 219 FE Jest; supplemented by 7 local verification addendum runs in feature/84-fds-final-qa-evidence: CORE-202 35 pass, Plan 03 14 pass/5 blocked, Phase B 3 pass/3 blocked, Phase C 5 pass/3 blocked, Phase D 8 pass, W04-B 6 pass/2 blocked, PERF-01..05 15/15 pass across 1,800 requests).

TC-FDS-001 | PASS (CI suite)

resolves a valid receptionist JWT

Requirement: FD-01 | Level: Unit | Priority: P1

Preconditions	JWT issuer/secret configured in isolated test
Input / steps	Valid signed receptionist bearer JWT
Expected result	Returns trusted receptionist principal; checks signature/issuer/expiry
Recorded actual result	Test assertions passed in named CI group; individual response payload not separately archived.
Test file / evidence	backend/src/security/auth/jwt-principal.resolver.spec.ts
Additional note	Does not prove manager-provisioned credential/login flow

TC-FDS-002 | NOT TESTED (integration)

Manager-provisioned receptionist login end-to-end

Requirement: FD-01 | Level: Cross-service E2E | Priority: P1

Preconditions	Live central authentication service
Input / steps	Manager provisions credentials; receptionist logs in
Expected result	Credentials provisioned; actual user can authenticate FDS
Recorded actual result	No verified execution evidence.
Test file / evidence	N/A - identity provisioning belongs to central auth/MAD integration
Additional note	FDS JWT authorization tests are not end-to-end account provisioning evidence

TC-FDS-003 | PASS (CI suite)

returns confirmed arrivals for the requested date

Requirement: FD-02 | Level: PostgreSQL integration | Priority: P1

Preconditions	Seeded bookings
Input / steps	Query arrivals for date
Expected result	Only confirmed arrivals for date returned
Recorded actual result	Test assertions passed in named CI group; individual response payload not separately archived.
Test file / evidence	backend/test/database/postgres-booking.repository.integration-spec.ts

TC-FDS-004 | PASS (CI suite)

returns checked-in departures for the requested date

Requirement: FD-02 | Level: PostgreSQL integration | Priority: P1

Preconditions	Seeded bookings
Input / steps	Query departures for date
Expected result	Only checked-in departures for date returned
Recorded actual result	Test assertions passed in named CI group; individual response payload not separately archived.
Test file / evidence	backend/test/database/postgres-booking.repository.integration-spec.ts

TC-FDS-005 | PASS (CI suite)

searches persisted bookings by guest name

Requirement: FD-03 | Level: PostgreSQL integration | Priority: P1

Preconditions	Seeded guest/booking
Input / steps	Search by guest name
Expected result	Returns matching persisted booking
Recorded actual result	Test assertions passed in named CI group; individual response payload not separately archived.
Test file / evidence	backend/test/database/postgres-booking.repository.integration-spec.ts

TC-FDS-006 | PASS (CI suite)

searches persisted bookings by NIC/passport

Requirement: FD-03 | Level: PostgreSQL integration | Priority: P1

Preconditions	Seeded guest/booking
Input / steps	Search by identity field
Expected result	Returns matching persisted booking
Recorded actual result	Test assertions passed in named CI group; individual response payload not separately archived.
Test file / evidence	backend/test/database/postgres-booking.repository.integration-spec.ts

TC-FDS-007 | PASS (CI suite)

creates a cash walk-in booking and completed payment

Requirement: FD-04 | Level: PostgreSQL integration | Priority: P1

Preconditions	Disposable database; valid room type
Input / steps	Create cash walk-in
Expected result	Booking and completed payment persisted in transaction
Recorded actual result	Test assertions passed in named CI group; individual response payload not separately archived.
Test file / evidence	backend/test/database/postgres-walk-in-booking.repository.integration-spec.ts

TC-FDS-008 | PASS (CI suite)

rolls back the booking when payment persistence fails

Requirement: FD-04 | Level: PostgreSQL integration | Priority: P1

Preconditions	Failure injected in payment stage
Input / steps	Create walk-in while payment persistence fails
Expected result	Booking is rolled back
Recorded actual result	Test assertions passed in named CI group; individual response payload not separately archived.
Test file / evidence	backend/test/database/postgres-walk-in-booking.repository.integration-spec.ts

TC-FDS-009 | PASS (CI suite)

checks in a confirmed booking with physical-document verification

Requirement: FD-05 | Level: PostgreSQL integration | Priority: P1

Preconditions	Confirmed booking, receptionist, vacant room
Input / steps	Physical-document verification on check-in
Expected result	Verification and check-in saved atomically
Recorded actual result	Test assertions passed in named CI group; individual response payload not separately archived.
Test file / evidence	backend/test/database/postgres-check-in.repository.integration-spec.ts

TC-FDS-010 | PASS (CI suite)

checks in a pre-assigned booking with scanned-copy metadata

Requirement: FD-05 | Level: PostgreSQL integration | Priority: P1

Preconditions	Confirmed booking, scanned-copy metadata
Input / steps	Check in with scanned document metadata
Expected result	Scanned-copy metadata persisted and booking checked in
Recorded actual result	Test assertions passed in named CI group; individual response payload not separately archived.
Test file / evidence	backend/test/database/postgres-check-in.repository.integration-spec.ts

TC-FDS-011 | PASS (CI suite)

rejects an occupied room

Requirement: FD-06 | Level: PostgreSQL integration | Priority: P1

Preconditions	Booked stay; occupied target room
Input / steps	Attempt room assignment on check-in
Expected result	Room conflict is rejected
Recorded actual result	Test assertions passed in named CI group; individual response payload not separately archived.
Test file / evidence	backend/test/database/postgres-check-in.repository.integration-spec.ts

TC-FDS-012 | PASS (CI suite)

checks in a confirmed booking with physical-document verification

Requirement: FD-06 | Level: PostgreSQL integration | Priority: P1

Preconditions	Confirmed booking, vacant compatible room
Input / steps	Check in and assign room
Expected result	Booking and room status updated
Recorded actual result	Test assertions passed in named CI group; individual response payload not separately archived.
Test file / evidence	backend/test/database/postgres-check-in.repository.integration-spec.ts

TC-FDS-013 | PASS (CI suite)

should check in and activate FOSS access after the domain transaction succeeds

Requirement: FD-07 | Level: Unit / mocked gateway | Priority: P1

Preconditions	Mock FOSS gateway injected
Input / steps	Successful check-in
Expected result	Service calls FOSS activation after successful domain operation
Recorded actual result	Test assertions passed in named CI group; individual response payload not separately archived.
Test file / evidence	backend/src/check-ins/check-in.service.spec.ts
Additional note	Mock is not proof of real FOSS integration

TC-FDS-014 | NOT TESTED (integration)

FDS-to-live-FOSS activation

Requirement: FD-07 | Level: Cross-service E2E | Priority: P1

Preconditions	Live FOSS contract & auth
Input / steps	Complete FDS check-in and inspect actual FOSS session
Expected result	FOSS session usable by checked-in guest
Recorded actual result	No verified execution evidence.
Test file / evidence	N/A - real FOSS cross-service test not present in reviewed FDS CI
Additional note	Mock activation is not production activation

TC-FDS-015 | PASS (CI suite)

should accept a registration-card print job

Requirement: FD-08 | Level: Unit / mock print | Priority: P1

Preconditions	Mock printer gateway
Input / steps	Request registration-card print
Expected result	Mock accepts print job
Recorded actual result	Test assertions passed in named CI group; individual response payload not separately archived.
Test file / evidence	backend/src/check-ins/gateways/mock-check-in-print.gateway.spec.ts
Additional note	No physical printer output demonstrated

TC-FDS-016 | NOT TESTED (hardware)

Registration card / receipt physically printed

Requirement: FD-08 | Level: Acceptance / hardware | Priority: P1

Preconditions	Printer device and real adapter connected
Input / steps	Request print and inspect paper output
Expected result	Physical registration card / receipt printed
Recorded actual result	No verified execution evidence.
Test file / evidence	N/A - real printer not integrated

TC-FDS-017 | PASS (CI suite)

atomically persists final payment, checked-out booking, cleaning room and checkout audit

Requirement: FD-09 | Level: PostgreSQL integration | Priority: P1

Preconditions	Checked-in booking; ready checkout
Input / steps	Persist checkout with settlement
Expected result	Payment, booking, room, audit remain consistent
Recorded actual result	Test assertions passed in named CI group; individual response payload not separately archived.
Test file / evidence	backend/test/database/postgres-checkout.repository.integration-spec.ts
Additional note	Payment gateway settlement is mocked separately

TC-FDS-018 | NOT TESTED (integration)

Real card/cash settlement at checkout

Requirement: FD-09 | Level: Payment integration | Priority: P1

Preconditions	Certified real payment adapter
Input / steps	Complete charge/refund/settlement
Expected result	Verified payment provider settlement
Recorded actual result	No verified execution evidence.
Test file / evidence	N/A - live payment adapter not evidenced

TC-FDS-019 | PASS (CI suite)

atomically persists final payment, checked-out booking, cleaning room and checkout audit

Requirement: FD-10 | Level: PostgreSQL integration | Priority: P1

Preconditions	Checked-in room, settled folio
Input / steps	Checkout booking
Expected result	Room marked for cleaning, booking checked out, audit persisted
Recorded actual result	Test assertions passed in named CI group; individual response payload not separately archived.
Test file / evidence	backend/test/database/postgres-checkout.repository.integration-spec.ts

TC-FDS-020 | NOT TESTED (integration)

Checkout-to-live-WKMS cleaning task and FOSS deactivation

Requirement: FD-10 | Level: Cross-service E2E | Priority: P1

Preconditions	Live RabbitMQ, WKMS, FOSS
Input / steps	Complete checkout; inspect WKMS and FOSS
Expected result	WKMS task created, FOSS access revoked
Recorded actual result	No verified execution evidence.
Test file / evidence	N/A - integrated checkout chain not verified

TC-FDS-021 | PASS (CI suite)

atomically reassigns a checked-in booking and audits the room change

Requirement: FD-11 | Level: PostgreSQL integration | Priority: P1

Preconditions	Checked-in booking, compatible new room
Input / steps	Change assigned room
Expected result	Booking, old/new rooms and audit update atomically
Recorded actual result	Test assertions passed in named CI group; individual response payload not separately archived.
Test file / evidence	backend/test/database/postgres-room-change.repository.integration-spec.ts

TC-FDS-022 | PASS (CI suite)

rejects an UNDER_MAINTENANCE target room

Requirement: FD-11 | Level: PostgreSQL integration | Priority: P1

Preconditions	Target room under maintenance
Input / steps	Try to move guest into target
Expected result	Unsafe room-change rejected
Recorded actual result	Test assertions passed in named CI group; individual response payload not separately archived.
Test file / evidence	backend/test/database/postgres-room-change.repository.integration-spec.ts

TC-FDS-023 | PASS (CI suite)

loads room-board data including all supported room states

Requirement: FD-12 | Level: PostgreSQL integration | Priority: P1

Preconditions	Seeded room inventory
Input / steps	GET room board snapshot
Expected result	Room-state snapshot includes supported states
Recorded actual result	Test assertions passed in named CI group; individual response payload not separately archived.
Test file / evidence	backend/test/database/postgres-room.repository.integration-spec.ts

TC-FDS-024 | PASS (CI suite)

allows a receptionist and delivers room events

Requirement: FD-12 | Level: Socket.IO E2E | Priority: P1

Preconditions	JWT receptionist; realtime gateway
Input / steps	Connect Socket.IO; publish room event
Expected result	Authenticated client receives room-status update
Recorded actual result	Test assertions passed in named CI group; individual response payload not separately archived.
Test file / evidence	backend/test/realtime.e2e-spec.ts

TC-FDS-025 | PASS (CI suite)

transitions a VACANT room to UNDER_MAINTENANCE and preserves lastClearedAt

Requirement: FD-13 | Level: PostgreSQL integration | Priority: P1

Preconditions	VACANT room
Input / steps	Block room for maintenance
Expected result	Status updates to UNDER_MAINTENANCE with required audit semantics
Recorded actual result	Test assertions passed in named CI group; individual response payload not separately archived.
Test file / evidence	backend/test/database/postgres-room.repository.integration-spec.ts

TC-FDS-026 | PASS (CI suite)

transitions UNDER_MAINTENANCE to VACANT

Requirement: FD-13 | Level: PostgreSQL integration | Priority: P1

Preconditions	Room under maintenance
Input / steps	Clear maintenance
Expected result	Status returns to VACANT
Recorded actual result	Test assertions passed in named CI group; individual response payload not separately archived.
Test file / evidence	backend/test/database/postgres-room.repository.integration-spec.ts

TC-FDS-027 | PASS (CI suite)

should create a WKMS task for a checked-in guest

Requirement: FD-14 | Level: Unit / mocked WKMS | Priority: P1

Preconditions	Checked-in guest; mocked WKMS
Input / steps	Request guest service task
Expected result	WKMS gateway method invoked with task payload
Recorded actual result	Test assertions passed in named CI group; individual response payload not separately archived.
Test file / evidence	backend/src/service-requests/service-requests.service.spec.ts
Additional note	No end-to-end WKMS execution proved

TC-FDS-028 | NOT TESTED (integration)

FDS-to-live-WKMS service request delivered

Requirement: FD-14 | Level: Cross-service E2E | Priority: P1

Preconditions	Both services deployed; API configured
Input / steps	Create FDS service request and inspect worker queue
Expected result	Worker task exists in WKMS
Recorded actual result	No verified execution evidence.
Test file / evidence	N/A - real WKMS flow not evidenced

TC-FDS-029 | PASS (CI suite)

loads the persisted checked-in booking context for a running folio

Requirement: FD-15 | Level: PostgreSQL integration | Priority: P1

Preconditions	Active stay and persisted booking total
Input / steps	Retrieve running folio
Expected result	Room charge based on authoritative booking data
Recorded actual result	Test assertions passed in named CI group; individual response payload not separately archived.
Test file / evidence	backend/test/database/postgres-folio.repository.integration-spec.ts

TC-FDS-030 | PASS (CI suite)

should return an empty external-charge list until live integration is connected

Requirement: FD-15 | Level: Unit / mock | Priority: P1

Preconditions	Mock external-charge gateway
Input / steps	Get external F&B/service charges
Expected result	Empty mock result, not fabricated charges
Recorded actual result	Test assertions passed in named CI group; individual response payload not separately archived.
Test file / evidence	backend/src/billing/gateways/mock-external-folio-charge.gateway.spec.ts
Additional note	Live kitchen/service charging not yet tested

TC-FDS-031 | PASS (CI suite)

rejects modification of an existing audit record

Requirement: FD-16 | Level: PostgreSQL integration | Priority: P1

Preconditions	Existing audit record + DB trigger
Input / steps	Attempt UPDATE audit_logs
Expected result	Database rejects mutation
Recorded actual result	Test assertions passed in named CI group; individual response payload not separately archived.
Test file / evidence	backend/test/database/audit-append-only.integration-spec.ts

TC-FDS-032 | PASS (CI suite)

rejects deletion of an existing audit record

Requirement: FD-16 | Level: PostgreSQL integration | Priority: P1

Preconditions	Existing audit record + DB trigger
Input / steps	Attempt DELETE audit_logs
Expected result	Database rejects deletion
Recorded actual result	Test assertions passed in named CI group; individual response payload not separately archived.
Test file / evidence	backend/test/database/audit-append-only.integration-spec.ts

TC-FDS-033 | PASS (CI suite)

allows a receptionist to read audit logs

Requirement: FD-16 | Level: HTTP E2E | Priority: P1

Preconditions	Receptionist bearer JWT
Input / steps	GET /audit-logs
Expected result	Authorized audit logs read
Recorded actual result	Test assertions passed in named CI group; individual response payload not separately archived.
Test file / evidence	backend/test/audit-rbac.e2e-spec.ts

TC-FDS-034 | PASS (CI suite)

should validate, push and ack a valid escalation

Requirement: FD-17 | Level: Unit / mocked broker | Priority: P1

Preconditions	Valid WKMS escalation fixture
Input / steps	Consume valid event
Expected result	Validate event; publish realtime notification; ack
Recorded actual result	Test assertions passed in named CI group; individual response payload not separately archived.
Test file / evidence	backend/src/realtime/task-escalation.consumer.spec.ts
Additional note	Unclaimed-after-15-min trigger requires WKMS evidence

TC-FDS-035 | NOT TESTED (integration)

Real WKMS 15-minute escalation policy

Requirement: FD-17 | Level: Cross-service integration | Priority: P1

Preconditions	Live WKMS task creation/timer
Input / steps	Leave task unclaimed for configured threshold
Expected result	WKMS generates proper escalation
Recorded actual result	No verified execution evidence.
Test file / evidence	N/A - WKMS timer test is outside FDS scope

TC-FDS-036 | PASS (CI suite)

should audit a successful manual assignment and remove its escalation

Requirement: FD-18 | Level: Unit / mocked WKMS | Priority: P1

Preconditions	Available worker returned by mock
Input / steps	Assign escalated task
Expected result	Assignment succeeds, audit written, escalation removed
Recorded actual result	Test assertions passed in named CI group; individual response payload not separately archived.
Test file / evidence	backend/src/service-requests/service-requests.service.spec.ts
Additional note	Test uses WKMS client mock

TC-FDS-037 | NOT TESTED (integration)

Real WKMS worker assignment

Requirement: FD-18 | Level: Cross-service E2E | Priority: P1

Preconditions	Live WKMS workers/assignment API
Input / steps	Assign task from FDS UI
Expected result	Correct real worker receives task
Recorded actual result	No verified execution evidence.
Test file / evidence	N/A - live WKMS mutation not verified
Additional note	Retained as NOT TESTED for production cross-service E2E; see FDS local verification addendum W04-B (6 PASS / 2 BLOCKED on local authenticated FDS-to-WKMS manual assignment).

TC-FDS-038 | PASS (CI suite)

should subscribe to the task escalation queue

Requirement: FD-19 | Level: Unit / mocked broker | Priority: P1

Preconditions	RabbitMQ consumer mocked
Input / steps	Subscribe to escalation queue
Expected result	Consumer wiring requests expected queue subscription
Recorded actual result	Test assertions passed in named CI group; individual response payload not separately archived.
Test file / evidence	backend/src/realtime/task-escalation.consumer.spec.ts

TC-FDS-039 | PASS (CI suite)

updates both indicators from realtime

Requirement: FD-19 | Level: Frontend unit/component | Priority: P1

Preconditions	Mock realtime escalation
Input / steps	Emit escalation update into component
Expected result	Visible indicators update
Recorded actual result	Test assertions passed in named CI group; individual response payload not separately archived.
Test file / evidence	frontend/src/components/layout/escalation-indicators.test.tsx

TC-FDS-040 | NOT TESTED (integration)

Live WKMS RabbitMQ event reaches receptionist UI

Requirement: FD-19 | Level: Cross-service E2E | Priority: P1

Preconditions	WKMS publisher, shared broker, FDS socket
Input / steps	Publish WKMS event; observe UI badge
Expected result	Notification and fallback consistent
Recorded actual result	No verified execution evidence.
Test file / evidence	N/A - live cross-subsystem scenario not verified
Additional note	Retained as NOT TESTED for production cross-service E2E; see FDS local verification addendum Phase B (3 PASS), Phase C (5 PASS), and Phase D (8 PASS browser realtime UX).

TC-FDS-041 | PASS (CI suite)

returns 403 when a worker reads bookings

Requirement: NFR-12 | Level: HTTP E2E | Priority: P1

Preconditions	Worker JWT
Input / steps	GET protected booking endpoint
Expected result	Forbidden for worker
Recorded actual result	Test assertions passed in named CI group; individual response payload not separately archived.
Test file / evidence	backend/test/bookings-rbac.e2e-spec.ts

TC-FDS-042 | PASS (CI suite)

rejects raw card properties before checkout service execution

Requirement: NFR-12 | Level: HTTP E2E | Priority: P1

Preconditions	Receptionist JWT
Input / steps	POST checkout with raw card property
Expected result	Request validation blocks card properties
Recorded actual result	Test assertions passed in named CI group; individual response payload not separately archived.
Test file / evidence	backend/test/folio-checkout-rbac.e2e-spec.ts

TC-FDS-043 | PASS (CI suite)

rejects a JWT with an invalid issuer

Requirement: NFR-12 | Level: Socket.IO E2E | Priority: P1

Preconditions	Wrong-issuer JWT
Input / steps	Connect socket with bad issuer
Expected result	Connection refused
Recorded actual result	Test assertions passed in named CI group; individual response payload not separately archived.
Test file / evidence	backend/test/realtime.e2e-spec.ts

5. Food Ordering and Service Requests (FOSS)

Repository: forever-hotel/food-ordering | Branch: develop | Tested revision: a69ba0a16ca941b783633427e5e31348bf681f17 (QA package commit: 9ac1e6f3df9f93ee76be6db479dc8c4bf6529328) | CI date: 8 November 2026

Evidence: https://github.com/forever-hotel/food-ordering/actions/runs/42490527447, https://github.com/forever-hotel/food-ordering/actions/runs/41833172660, documents/testing/test-register-source.csv

Register entries: 54; CI-suite-linked pass: 43; not tested/not verified: 11 (9 cross-service/architecture/load NOT TESTED, 2 headless browser NOT VERIFIED). Automated CI passes 558 backend unit tests, 237 backend E2E tests (5 skipped), and 269 frontend Node unit tests (1 skipped).

TC-FOSS-001 | PASS (CI suite)

Authenticates valid QR code slip with verified booking and room

Requirement: GA-01 | Level: E2E Integration | Priority: P1

Preconditions	Active checked-in booking in FDS, valid QR slip payload
Input / steps	POST /guest-sessions/qr with valid code and room 101
Expected result	Issues signed Bearer session token with 24h TTL and room context
Recorded actual result	Test assertions passed in CI suite; session returned with 200 OK.
Test file / evidence	backend/test/qr-activation-lifecycle.e2e-spec.ts
Additional note	Validates core QR guest onboarding journey

TC-FOSS-002 | PASS (CI suite)

Rejects tampered or forged QR activation codes

Requirement: GA-01 | Level: Unit | Priority: P1

Preconditions	Configured QR signing HMAC secret
Input / steps	Redeem forged access code with altered signature
Expected result	Rejection with 401 Unauthorized; no session created
Recorded actual result	Test assertions passed in CI suite; throws InvalidQrSignatureException.
Test file / evidence	backend/src/modules/guest-sessions/services/qr-activation.service.spec.ts
Additional note	Prevents unauthorized guest impersonation

TC-FOSS-003 | PASS (CI suite)

Internal FDS check-in triggers FOSS QR slip issuance

Requirement: FD-07 | Level: Unit / Controller | Priority: P1

Preconditions	FDS internal service authorization key configured
Input / steps	POST /guest-sessions/internal/qr/issue with bookingId and roomNumber
Expected result	Generates high-entropy access code and printable QR slip payload
Recorded actual result	Test assertions passed in CI suite; returns 201 Created with QR payload.
Test file / evidence	backend/src/modules/guest-sessions/guest-sessions.controller.spec.ts
Additional note	FDS-to-FOSS integration boundary

TC-FOSS-004 | NOT TESTED (integration)

Live FDS check-in to FOSS session activation

Requirement: FD-07 | Level: Cross-service E2E | Priority: P1

Preconditions	Deployed FDS and FOSS instances with shared network
Input / steps	Perform check-in in FDS and scan issued slip in FOSS
Expected result	FOSS guest session active and bound to physical room
Recorded actual result	No verified execution evidence.
Test file / evidence	N/A - live FDS cross-service test not present in CI
Additional note	Simulated via MockBookingAdapter in development

TC-FOSS-005 | PASS (CI suite)

Validates active guest session token via me endpoint

Requirement: GA-01 | Level: E2E Integration | Priority: P1

Preconditions	Active guest session Bearer token
Input / steps	GET /guest-sessions/me with Authorization header
Expected result	Returns 200 OK with bookingId roomNumber and expiresAt
Recorded actual result	Test assertions passed in CI suite; guest identity returned.
Test file / evidence	backend/test/guest-sessions.e2e-spec.ts
Additional note	Session introspection contract

TC-FOSS-006 | PASS (CI suite)

Rejects expired or malformed Bearer tokens

Requirement: GA-01 | Level: E2E Integration | Priority: P1

Preconditions	None
Input / steps	GET /guest-sessions/me with expired token or invalid signature
Expected result	Returns 401 Unauthorized with standard error envelope
Recorded actual result	Test assertions passed in CI suite; 401 returned.
Test file / evidence	backend/test/room-scoped-auth.e2e-spec.ts
Additional note	Enforces session TTL bounds

TC-FOSS-007 | PASS (CI suite)

Revokes guest session immediately upon FDS checkout event

Requirement: GA-01b | Level: E2E Integration | Priority: P1

Preconditions	Active guest session for Room 101
Input / steps	POST /guest-sessions/internal/events/checkout for Room 101
Expected result	Session marked REVOKED; subsequent guest requests return 401
Recorded actual result	Test assertions passed in CI suite; session revoked and disconnected.
Test file / evidence	backend/test/fds-checkout-session-revocation.e2e-spec.ts
Additional note	Prevents post-checkout ordering

TC-FOSS-008 | NOT TESTED (integration)

Live FDS checkout to FOSS session revocation

Requirement: FD-10 | Level: Cross-service E2E | Priority: P1

Preconditions	Live FDS instance and RabbitMQ broker
Input / steps	Complete checkout in FDS and test active FOSS client
Expected result	FOSS app immediately terminates session and redirects to scan
Recorded actual result	No verified execution evidence.
Test file / evidence	N/A - live checkout event propagation not executed
Additional note	Verified in local mock and E2E event simulation

TC-FOSS-009 | PASS (CI suite)

Enforces strict room ownership on room-specific endpoints

Requirement: GA-11 | Level: E2E Integration | Priority: P1

Preconditions	Active session token for Room 101
Input / steps	GET /guest-sessions/rooms/102 with Room 101 token
Expected result	Returns 403 Forbidden preventing cross-room IDOR access
Recorded actual result	Test assertions passed in CI suite; RoomOwnershipGuard denies access.
Test file / evidence	backend/test/room-scoped-auth.e2e-spec.ts
Additional note	Prevents room hopping attacks

TC-FOSS-010 | PASS (CI suite)

Enforces booking isolation on booking-specific endpoints

Requirement: GA-11 | Level: E2E Integration | Priority: P1

Preconditions	Active session token for Booking BKG-001
Input / steps	GET /guest-sessions/bookings/BKG-002 with BKG-001 token
Expected result	Returns 403 Forbidden preventing cross-booking IDOR
Recorded actual result	Test assertions passed in CI suite; RoomOwnershipGuard denies access.
Test file / evidence	backend/test/room-scoped-auth.e2e-spec.ts
Additional note	Booking partition isolation

TC-FOSS-011 | PASS (CI suite)

Frontend partitions session storage between Demo and API modes

Requirement: NFR-12 | Level: Frontend Unit | Priority: P1

Preconditions	Browser localStorage mock
Input / steps	Switch between DATA_MODE demo and api
Expected result	Session tokens stored in separate keys; previous mode cleared
Recorded actual result	Test assertions passed in node test runner; tokens partitioned cleanly.
Test file / evidence	frontend/tests/auth-lifecycle-isolation.test.ts
Additional note	Eliminates state contamination (#101)

TC-FOSS-012 | PASS (CI suite)

Frontend rejects demo tokens when running in API mode

Requirement: NFR-12 | Level: Frontend Unit | Priority: P1

Preconditions	Demo mode token in storage with NEXT_PUBLIC_DATA_MODE=api
Input / steps	Attempt API call with demo token
Expected result	Returns 401 Unauthorized; redirects to activation
Recorded actual result	Test assertions passed in node test runner; demo token rejected.
Test file / evidence	frontend/tests/auth-lifecycle-isolation.test.ts
Additional note	Prevents demo bypass in staging/production

TC-FOSS-013 | PASS (CI suite)

Retrieves categorized menu catalog with Colombo meal windows

Requirement: GA-02 | Level: E2E Integration | Priority: P1

Preconditions	Authenticated guest session
Input / steps	GET /menu
Expected result	Returns categories items pricing allergens and active meal window
Recorded actual result	Test assertions passed in CI suite; 200 OK with categorized items.
Test file / evidence	backend/test/menu.e2e-spec.ts
Additional note	Authoritative menu schema

TC-FOSS-014 | PASS (CI suite)

Determines active meal periods based on Colombo local time

Requirement: GA-02 | Level: Unit / Service | Priority: P1

Preconditions	System clock set to 13:00 Colombo time
Input / steps	Invoke determineActiveMealPeriods()
Expected result	Returns LUNCH and ALL_DAY meal periods as active
Recorded actual result	Test assertions passed in CI suite; correct periods returned.
Test file / evidence	backend/src/modules/menu/menu.service.spec.ts
Additional note	Sri Lanka timezone awareness

TC-FOSS-015 | PASS (CI suite)

Enforces meal window cutoff for scheduled categories

Requirement: GA-06 | Level: E2E Integration | Priority: P1

Preconditions	System clock set outside breakfast hours (11:30 AM)
Input / steps	GET /menu/windows and inspect BREAKFAST category
Expected result	Breakfast items flagged unavailable with cutoff warning
Recorded actual result	Test assertions passed in CI suite; cutoff status accurately reported.
Test file / evidence	backend/test/menu.e2e-spec.ts
Additional note	Meal period cutoff enforcement

TC-FOSS-016 | PASS (CI suite)

Frontend displays category-specific meal window cutoff warnings

Requirement: GA-06 | Level: Frontend Unit | Priority: P1

Preconditions	Menu catalog with Breakfast ending at 10:30 AM
Input / steps	Render menu item list with active time remaining
Expected result	Displays hospitality phrasing and warning banner for closing window
Recorded actual result	Test assertions passed in node test runner; warning rendered.
Test file / evidence	frontend/tests/meal-windows-and-messaging.test.ts
Additional note	FDR Figure 33 fidelity

TC-FOSS-017 | NOT TESTED (integration)

Live KMS menu catalog synchronization

Requirement: GA-02 | Level: Cross-service Integration | Priority: P1

Preconditions	Live KMS instance with menu management API
Input / steps	GET /menu with real KMS adapter enabled
Expected result	FOSS menu reflects KMS published items and prices
Recorded actual result	No verified execution evidence.
Test file / evidence	N/A - live KMS endpoint not connected in CI
Additional note	Simulated via MockKmsAdapter in development

TC-FOSS-018 | PASS (CI suite)

Retrieves complete item details including allergens and calories

Requirement: GA-03 | Level: E2E Integration | Priority: P1

Preconditions	Valid item ID in menu catalog
Input / steps	GET /menu/items/:id for Grilled Chicken Rice
Expected result	Returns calories (650 kcal) allergens portion and price (LKR 1850)
Recorded actual result	Test assertions passed in CI suite; detailed item schema verified.
Test file / evidence	backend/test/menu.e2e-spec.ts
Additional note	FDR Figure 32 fidelity

TC-FOSS-019 | PASS (CI suite)

Enforces maximum cart portion limit of 20 items per line

Requirement: GA-04 | Level: Frontend Unit | Priority: P1

Preconditions	Active cart state
Input / steps	Attempt to add 21 units of a single menu item
Expected result	Rejects quantity 21; clamps maximum quantity to 20
Recorded actual result	Test assertions passed in node test runner; rejects quantity > 20.
Test file / evidence	frontend/tests/pricing-and-quantity.test.ts
Additional note	Anti-exhaustion portion cap

TC-FOSS-020 | PASS (CI suite)

Rejects non-positive cart quantities (quantity 0 or negative)

Requirement: GA-04 | Level: Frontend Unit | Priority: P1

Preconditions	Active cart state
Input / steps	Submit cart item with quantity 0 or -1
Expected result	Validation error; item removed or rejected
Recorded actual result	Test assertions passed in node test runner; rejects non-positive qty.
Test file / evidence	frontend/tests/pricing-and-quantity.test.ts
Additional note	Boundary validation

TC-FOSS-021 | PASS (CI suite)

Preserves allergy and dietary preferences note up to 300 chars

Requirement: GA-05 | Level: Frontend Unit | Priority: P1

Preconditions	Active cart with 2 items
Input / steps	Add allergy note of 250 characters (e.g. No shellfish extra spicy)
Expected result	Note persisted in cart state and included in order payload
Recorded actual result	Test assertions passed in node test runner; note correctly saved.
Test file / evidence	frontend/tests/cart.test.ts
Additional note	SRS GA-05 compliance

TC-FOSS-022 | PASS (CI suite)

Rejects allergy and dietary note exceeding 300 character limit

Requirement: GA-05 | Level: E2E Integration | Priority: P1

Preconditions	Authenticated guest session
Input / steps	POST /food-orders with allergyNote of 301 characters
Expected result	Returns 400 Bad Request with allergyNote length error
Recorded actual result	Test assertions passed in CI suite; validation error returned.
Test file / evidence	backend/test/api-validation-and-errors.e2e-spec.ts
Additional note	Input boundary enforcement

TC-FOSS-023 | PASS (CI suite)

Calculates monetary totals with integer minor units precision

Requirement: GA-04 | Level: Frontend Unit | Priority: P1

Preconditions	Cart with fractional prices (e.g. LKR 1850.50)
Input / steps	Calculate subtotal and total across multiple items
Expected result	Exact integer minor unit precision without IEEE-754 drift
Recorded actual result	Test assertions passed in node test runner; zero floating point drift.
Test file / evidence	frontend/tests/pricing-and-quantity.test.ts
Additional note	Monetary rounding integrity

TC-FOSS-024 | PASS (CI suite)

Submits food order atomically with server-calculated pricing

Requirement: GA-08 | Level: E2E Integration | Priority: P1

Preconditions	Authenticated session for Room 101
Input / steps	POST /food-orders with items allergyNote and idempotencyKey
Expected result	Order created with status SUBMITTED total LKR 3700 and ETA 30 min
Recorded actual result	Test assertions passed in CI suite; returns 201 Created.
Test file / evidence	backend/test/food-orders.e2e-spec.ts
Additional note	Core order submission flow

TC-FOSS-025 | PASS (CI suite)

Idempotent order placement prevents double-billing on replay

Requirement: NFR-05 | Level: E2E Integration | Priority: P1

Preconditions	Submitted order with Idempotency-Key idemp-ord-123
Input / steps	Replay identical POST /food-orders with same idempotency key
Expected result	Returns original order response without creating duplicate order
Recorded actual result	Test assertions passed in CI suite; identical order returned.
Test file / evidence	backend/test/food-orders.e2e-spec.ts
Additional note	Double-tap protection

TC-FOSS-026 | PASS (CI suite)

Calculates estimated delivery time window on confirmation

Requirement: GA-08 | Level: Frontend Unit | Priority: P1

Preconditions	Order placed at 12:00 with base prep time 25 min
Input / steps	Render order confirmation screen (Figure 35)
Expected result	Displays ETA range (25-35 minutes) and delivery countdown
Recorded actual result	Test assertions passed in node test runner; ETA formatted correctly.
Test file / evidence	frontend/tests/orders.test.ts
Additional note	FDR Figure 35 fidelity

TC-FOSS-027 | PASS (CI suite)

Dispatches order placement notification to kitchen via RabbitMQ

Requirement: KM-06 | Level: Integration | Priority: P1

Preconditions	Running RabbitMQ broker and order created
Input / steps	Inspect published messages on forever.hotel.events exchange
Expected result	Message published with routing key foss.order.placed and HMAC signature
Recorded actual result	Test assertions passed in CI suite; signed envelope verified.
Test file / evidence	backend/test/rabbitmq-live-integration.e2e-spec.ts
Additional note	KMS integration event

TC-FOSS-028 | PASS (CI suite)

Retrieves active stay order history for authenticated room

Requirement: GA-09 | Level: E2E Integration | Priority: P1

Preconditions	2 completed orders and 1 active order for Room 101
Input / steps	GET /food-orders with Room 101 token
Expected result	Returns array of 3 orders scoped strictly to current stay
Recorded actual result	Test assertions passed in CI suite; all 3 orders returned.
Test file / evidence	backend/test/food-orders.e2e-spec.ts
Additional note	Stay order history

TC-FOSS-029 | PASS (CI suite)

Filters out orders belonging to other rooms or previous stays

Requirement: GA-09 | Level: E2E Integration | Priority: P1

Preconditions	Orders seeded for Room 101 and Room 102
Input / steps	GET /food-orders with Room 101 token
Expected result	Returns only Room 101 orders; zero Room 102 orders leaked
Recorded actual result	Test assertions passed in CI suite; strict room scoping verified.
Test file / evidence	backend/test/food-orders.e2e-spec.ts
Additional note	Privacy isolation

TC-FOSS-030 | PASS (CI suite)

Allows order cancellation within 10-minute grace window

Requirement: GA-07 | Level: E2E Integration | Priority: P1

Preconditions	Order placed 3 minutes ago in SUBMITTED status
Input / steps	POST /food-orders/:id/cancel with Room 101 token
Expected result	Order status transitioned to CANCELLED; returns 200 OK
Recorded actual result	Test assertions passed in CI suite; order successfully cancelled.
Test file / evidence	backend/test/food-orders.e2e-spec.ts
Additional note	10-minute cancellation rule

TC-FOSS-031 | PASS (CI suite)

Rejects order cancellation after 10-minute cutoff

Requirement: GA-07 | Level: E2E Integration | Priority: P1

Preconditions	Order placed 11 minutes ago
Input / steps	POST /food-orders/:id/cancel
Expected result	Returns 409 Conflict with cancellation window expired message
Recorded actual result	Test assertions passed in CI suite; 409 returned.
Test file / evidence	backend/test/food-orders.e2e-spec.ts
Additional note	Cancellation deadline guard

TC-FOSS-032 | PASS (CI suite)

Rejects order cancellation once kitchen advances status to PREPARING

Requirement: GA-07 | Level: E2E Integration | Priority: P1

Preconditions	Order placed 4 minutes ago but kitchen advanced to PREPARING
Input / steps	POST /food-orders/:id/cancel
Expected result	Returns 409 Conflict (cannot cancel once preparation has begun)
Recorded actual result	Test assertions passed in CI suite; 409 returned.
Test file / evidence	backend/test/food-orders.e2e-spec.ts
Additional note	Kitchen lock-in rule

TC-FOSS-033 | PASS (CI suite)

Creates service request for one of 6 canonical categories

Requirement: GA-10 | Level: E2E Integration | Priority: P1

Preconditions	Authenticated session for Room 101
Input / steps	POST /service-requests with category EXTRA_TOWELS and description
Expected result	Returns 201 Created with status SUBMITTED and assigned timeline
Recorded actual result	Test assertions passed in CI suite; request persisted.
Test file / evidence	backend/test/service-requests.e2e-spec.ts
Additional note	Service request lifecycle

TC-FOSS-034 | PASS (CI suite)

Auto-binds room number from active session to service request

Requirement: GA-11 | Level: E2E Integration | Priority: P1

Preconditions	Room 101 session token
Input / steps	POST /service-requests without roomNumber in body
Expected result	Backend automatically populates room 101 from verified JWT principal
Recorded actual result	Test assertions passed in CI suite; roomNumber set to 101.
Test file / evidence	backend/test/service-requests.e2e-spec.ts
Additional note	Tamper-proof room binding

TC-FOSS-035 | PASS (CI suite)

Tracks service request progression Submitted to Completed

Requirement: GA-12 | Level: E2E Integration | Priority: P1

Preconditions	Service request in ASSIGNED status
Input / steps	Receive WKMS task status event COMPLETED
Expected result	Request status transitions to COMPLETED and audits completion time
Recorded actual result	Test assertions passed in CI suite; status updated to COMPLETED.
Test file / evidence	backend/test/service-requests.e2e-spec.ts
Additional note	WKMS task reconciliation

TC-FOSS-036 | NOT TESTED (integration)

Live WKMS worker task dispatch and delivery

Requirement: WM-01c | Level: Cross-service Integration | Priority: P1

Preconditions	Live WKMS instance and mobile worker app
Input / steps	Submit service request in FOSS and assign worker in WKMS
Expected result	Worker receives task; guest UI updates on worker progress
Recorded actual result	No verified execution evidence.
Test file / evidence	N/A - live WKMS worker dispatch not connected in CI
Additional note	Simulated via MockWkmsAdapter

TC-FOSS-037 | PASS (CI suite)

Submits guest complaint with category severity and description

Requirement: GA-13 | Level: E2E Integration | Priority: P1

Preconditions	Authenticated session for Room 101
Input / steps	POST /complaints with category FOOD_QUALITY severity HIGH and desc
Expected result	Returns 201 Created with complaint reference and status SUBMITTED
Recorded actual result	Test assertions passed in CI suite; complaint created.
Test file / evidence	backend/test/complaints.e2e-spec.ts
Additional note	Complaint intake

TC-FOSS-038 | PASS (CI suite)

Assigns unique high-entropy CMP reference identifier

Requirement: GA-14 | Level: E2E Integration | Priority: P1

Preconditions	New complaint submission
Input / steps	POST /complaints
Expected result	Returns unique reference matching CMP-YYYY-XXXX pattern
Recorded actual result	Test assertions passed in CI suite; CMP reference generated.
Test file / evidence	backend/test/complaints.e2e-spec.ts
Additional note	Reference traceability

TC-FOSS-039 | PASS (CI suite)

Retrieves stay complaints list and tracks manager resolution

Requirement: GA-15 | Level: E2E Integration | Priority: P1

Preconditions	Active stay complaint submitted
Input / steps	GET /complaints with Room 101 token
Expected result	Returns array containing submitted complaint and current status
Recorded actual result	Test assertions passed in CI suite; complaints returned.
Test file / evidence	backend/test/complaints.e2e-spec.ts
Additional note	Stay complaint tracking

TC-FOSS-040 | NOT TESTED (integration)

Live MAD manager dashboard complaint resolution

Requirement: GA-15 | Level: Cross-service Integration | Priority: P1

Preconditions	Live MAD instance with manager dashboard
Input / steps	Submit complaint in FOSS and resolve in MAD dashboard
Expected result	FOSS complaint status reflects resolution and manager notes
Recorded actual result	No verified execution evidence.
Test file / evidence	N/A - live MAD manager dashboard not connected in CI
Additional note	Simulated via MockMadComplaintAdapter

TC-FOSS-041 | PASS (CI suite)

Streams real-time events via Server-Sent Events with Bearer auth

Requirement: NFR-03 | Level: E2E Integration | Priority: P1

Preconditions	Authenticated guest session token
Input / steps	GET /realtime/events/stream with Authorization: Bearer header
Expected result	Opens text/event-stream and sends keepalive ping
Recorded actual result	Test assertions passed in CI suite; SSE stream connected.
Test file / evidence	backend/test/realtime.e2e-spec.ts
Additional note	SSE realtime feed

TC-FOSS-042 | PASS (CI suite)

Filters realtime events strictly to authenticated room number

Requirement: NFR-03 | Level: Frontend Unit | Priority: P1

Preconditions	Connected SSE stream for Room 101
Input / steps	Emit order status event for Room 102 across stream
Expected result	Room 101 client discards Room 102 event; zero data leakage
Recorded actual result	Test assertions passed in node test runner; event filtered.
Test file / evidence	frontend/tests/realtime-client-security.test.ts
Additional note	Room data isolation

TC-FOSS-043 | PASS (CI suite)

Falls back to polling when SSE connection fails

Requirement: NFR-03 | Level: Frontend Unit | Priority: P1

Preconditions	Network error on SSE endpoint (503 Service Unavailable)
Input / steps	Realtime client detects SSE disconnect
Expected result	Switches transparently to GET /realtime/events/poll at 5s intervals
Recorded actual result	Test assertions passed in node test runner; polling initiated.
Test file / evidence	frontend/tests/realtime-client-security.test.ts
Additional note	Resilient polling fallback

TC-FOSS-044 | PASS (CI suite)

Disconnects realtime client immediately upon SESSION_REVOKED

Requirement: NFR-14 | Level: Frontend Unit | Priority: P1

Preconditions	Active SSE stream for Room 101
Input / steps	Receive SESSION_REVOKED event
Expected result	Client closes SSE stream and clears active credentials
Recorded actual result	Test assertions passed in node test runner; clean disconnection.
Test file / evidence	frontend/tests/realtime-client-security.test.ts
Additional note	Session termination

TC-FOSS-045 | PASS (CI suite)

Processes signed RabbitMQ messages from KMS and WKMS

Requirement: NFR-05 | Level: Integration | Priority: P1

Preconditions	Running RabbitMQ broker and valid signed message
Input / steps	Publish message with valid HMAC to forever.hotel.events exchange
Expected result	Consumer validates signature and updates domain entity
Recorded actual result	Test assertions passed in CI suite; message consumed.
Test file / evidence	backend/test/rabbitmq-live-integration.e2e-spec.ts
Additional note	Signed event consumption

TC-FOSS-046 | PASS (CI suite)

Discards duplicate RabbitMQ messages using event deduplication

Requirement: NFR-05 | Level: Unit / Messaging | Priority: P1

Preconditions	Processed event ID recorded in cache
Input / steps	Deliver second message with identical event ID
Expected result	Consumer detects duplicate logs warning and acknowledges message
Recorded actual result	Test assertions passed in CI suite; duplicate discarded cleanly.
Test file / evidence	backend/src/messaging/rabbitmq-consumer.service.spec.ts
Additional note	Idempotent consumer

TC-FOSS-047 | PASS (CI suite)

Routes unparseable or poisoned messages to Dead-Letter Queue

Requirement: NFR-05 | Level: Unit / Messaging | Priority: P1

Preconditions	Malformed JSON payload without headers
Input / steps	Deliver message to queue
Expected result	Consumer nacks with requeue=false; broker routes to forever.hotel.dlq
Recorded actual result	Test assertions passed in CI suite; routed to DLQ.
Test file / evidence	backend/src/messaging/rabbitmq-consumer.service.spec.ts
Additional note	Poison message isolation

TC-FOSS-048 | NOT TESTED (architecture)

Transactional outbox consistency across DB and RabbitMQ

Requirement: NFR-05 | Level: Architectural Integration | Priority: P1

Preconditions	PostgreSQL transaction and broker outage
Input / steps	Write order to DB while RabbitMQ is unreachable
Expected result	Events buffered in outbox table and delivered upon broker recovery
Recorded actual result	No verified execution evidence.
Test file / evidence	N/A - transactional outbox not implemented
Additional note	Outbox not implemented; CR-outbox-foss.md proposed

TC-FOSS-049 | PASS (CI suite)

Operates transparently when optional Redis cache is disabled

Requirement: NFR-01 | Level: Integration | Priority: P1

Preconditions	REDIS_ENABLED=false (default setting)
Input / steps	Invoke menu catalog and session lookups
Expected result	Service falls back directly to database without throwing errors
Recorded actual result	Test assertions passed in CI suite; transparent DB fallback verified.
Test file / evidence	backend/test/redis-integration.e2e-spec.ts
Additional note	Non-authoritative cache safety

TC-FOSS-050 | NOT TESTED (load)

API latency p95 under 500ms at 100 concurrent users

Requirement: NFR-01 | Level: Load Performance | Priority: P1

Preconditions	Isolated staging environment with 100 virtual users
Input / steps	Execute k6 5-minute ramp load against menu and ordering APIs
Expected result	95% of HTTP responses complete in under 500 ms
Recorded actual result	No verified execution evidence.
Test file / evidence	N/A - no load test executed
Additional note	Load tests not executed against shared DB

TC-FOSS-051 | NOT TESTED (performance)

End-to-end event propagation latency under 1 second

Requirement: NFR-03 | Level: Performance | Priority: P1

Preconditions	Deployed KMS and FOSS with shared RabbitMQ broker
Input / steps	Update order status in KMS and measure SSE receipt in FOSS
Expected result	Status update arrives in browser within 1000 ms
Recorded actual result	No verified execution evidence.
Test file / evidence	N/A - integrated benchmark not executed
Additional note	Requires live multi-service benchmark

TC-FOSS-052 | NOT TESTED (load)

Supports 300 concurrent SSE connections without memory exhaustion

Requirement: NFR-14 | Level: Load Performance | Priority: P1

Preconditions	Staging NestJS backend with 300 active rooms
Input / steps	Connect 300 simultaneous SSE clients and hold for 30 minutes
Expected result	Server maintains connections with stable heap memory (< 256 MB)
Recorded actual result	No verified execution evidence.
Test file / evidence	N/A - connection soak test not executed
Additional note	Planned scenario for staging cluster

TC-FOSS-053 | NOT VERIFIED (browser)

Automated Playwright E2E browser guest ordering journey

Requirement: NFR-12 | Level: Browser E2E | Priority: P1

Preconditions	Running Next.js frontend and NestJS backend on host network
Input / steps	Automated Playwright browser executes scan cart and order flow
Expected result	Order placed successfully and confirmation displayed in browser
Recorded actual result	Sandbox socket isolation prevents headless execution.
Test file / evidence	frontend/tests/e2e/menu-cart-ordering.spec.ts
Additional note	Loopback socket constraint in sandbox

TC-FOSS-054 | NOT VERIFIED (browser)

Automated Axe-core WCAG 2.1 AA accessibility audit

Requirement: GA-03 | Level: Accessibility | Priority: P2

Preconditions	Playwright browser visiting all 4 main tabs
Input / steps	Run AxeBuilder scan for WCAG 2.1 AA contrast and aria labels
Expected result	Zero critical or serious accessibility violations reported
Recorded actual result	Headless browser audit pending host execution.
Test file / evidence	frontend/tests/e2e/accessibility-wcag-axe.spec.ts
Additional note	Authored in test suite but unverified in CI

6. Kitchen Management System (KMS)

Repository: forever-hotel/kitchen-management | Branch: develop | Tested revision: 1f56edd2ff1166c61d34daae3bdcc6809ab9ef02 (Oct 10 register baseline: 89087002ac2df8f07b6140c00111979c75bc2799) | CI date: 8 November 2026

Evidence: https://github.com/forever-hotel/kitchen-management/actions/runs/42502180570, https://github.com/forever-hotel/kitchen-management/actions/runs/38038213541

Register entries: 82; CI-suite-linked pass: 76; not tested/not verified: 6. By 8 November 2026 (commit 1f56edd2ff, Run 42502180570), KMS automated CI expanded from 76 to 665 passing tests (234 backend unit across 54 suites, 58 backend E2E across 10 passing suites with 6 skipped, and 373 frontend Jest tests across 59 suites).

TC-KMS-APP-001 | PASS (CI suite)

AppController responds with the foundational Hello World message

Requirement: Not recorded | Area: Foundations | Level: Unit | Priority: Not recorded | Kind: Automated

Preconditions	Test fixture/mocks as defined by the source file
Input / steps	Execute the named automated assertion in the source file
Expected result	AppController responds with the foundational Hello World message
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	backend/src/app.controller.spec.ts Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/backend/src/app.controller.spec.ts

TC-ENV-001 | PASS (CI suite)

Required environment values use approved development defaults

Requirement: Not recorded | Area: Configuration | Level: Unit | Priority: Not recorded | Kind: Automated

Preconditions	Test fixture/mocks as defined by the source file
Input / steps	Execute the named automated assertion in the source file
Expected result	Required environment values use approved development defaults
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	backend/src/config/validation.schema.spec.ts Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/backend/src/config/validation.schema.spec.ts

TC-ENV-002 | PASS (CI suite)

Invalid port configuration is rejected

Requirement: Not recorded | Area: Configuration | Level: Unit | Priority: Not recorded | Kind: Automated

Preconditions	Test fixture/mocks as defined by the source file
Input / steps	Execute the named automated assertion in the source file
Expected result	Invalid port configuration is rejected
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	backend/src/config/validation.schema.spec.ts Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/backend/src/config/validation.schema.spec.ts

TC-ENV-003 | PASS (CI suite)

Unsupported runtime environment is rejected

Requirement: Not recorded | Area: Configuration | Level: Unit | Priority: Not recorded | Kind: Automated

Preconditions	Test fixture/mocks as defined by the source file
Input / steps	Execute the named automated assertion in the source file
Expected result	Unsupported runtime environment is rejected
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	backend/src/config/validation.schema.spec.ts Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/backend/src/config/validation.schema.spec.ts

TC-KMS-DB-001 | PASS (CI suite)

Local PostgreSQL disables automatic synchronization and SSL

Requirement: Not recorded | Area: Database configuration | Level: Unit | Priority: Not recorded | Kind: Automated

Preconditions	Test fixture/mocks as defined by the source file
Input / steps	Execute the named automated assertion in the source file
Expected result	Local PostgreSQL disables automatic synchronization and SSL
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	backend/src/config/database.config.spec.ts Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/backend/src/config/database.config.spec.ts

TC-KMS-DB-002 | PASS (CI suite)

Managed PostgreSQL enables SSL with certificate verification

Requirement: Not recorded | Area: Database configuration | Level: Unit | Priority: Not recorded | Kind: Automated

Preconditions	Test fixture/mocks as defined by the source file
Input / steps	Execute the named automated assertion in the source file
Expected result	Managed PostgreSQL enables SSL with certificate verification
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	backend/src/config/database.config.spec.ts Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/backend/src/config/database.config.spec.ts

TC-KMS-DB-003 | PASS (CI suite)

Missing DATABASE_URL fails validation

Requirement: Not recorded | Area: Database configuration | Level: Unit | Priority: Not recorded | Kind: Automated

Preconditions	Test fixture/mocks as defined by the source file
Input / steps	Execute the named automated assertion in the source file
Expected result	Missing DATABASE_URL fails validation
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	backend/src/config/validation.schema.spec.ts Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/backend/src/config/validation.schema.spec.ts

TC-KMS-DB-004 | PASS (CI suite)

Unsupported DATABASE_URL schemes fail validation

Requirement: Not recorded | Area: Database configuration | Level: Unit | Priority: Not recorded | Kind: Automated

Preconditions	Test fixture/mocks as defined by the source file
Input / steps	Execute the named automated assertion in the source file
Expected result	Unsupported DATABASE_URL schemes fail validation
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	backend/src/config/validation.schema.spec.ts Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/backend/src/config/validation.schema.spec.ts

TC-KMS-DB-005 | PASS (CI suite)

Initialized responsive database returns ready

Requirement: Not recorded | Area: Database health | Level: Unit | Priority: Not recorded | Kind: Automated

Preconditions	Test fixture/mocks as defined by the source file
Input / steps	Execute the named automated assertion in the source file
Expected result	Initialized responsive database returns ready
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	backend/src/database/database-health.service.spec.ts Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/backend/src/database/database-health.service.spec.ts

TC-KMS-DB-006 | PASS (CI suite)

Uninitialized database returns not ready

Requirement: Not recorded | Area: Database health | Level: Unit | Priority: Not recorded | Kind: Automated

Preconditions	Test fixture/mocks as defined by the source file
Input / steps	Execute the named automated assertion in the source file
Expected result	Uninitialized database returns not ready
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	backend/src/database/database-health.service.spec.ts Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/backend/src/database/database-health.service.spec.ts

TC-KMS-DB-007 | PASS (CI suite)

Failed database query returns not ready without leaking the error

Requirement: Not recorded | Area: Database health | Level: Unit | Priority: Not recorded | Kind: Automated

Preconditions	Test fixture/mocks as defined by the source file
Input / steps	Execute the named automated assertion in the source file
Expected result	Failed database query returns not ready without leaking the error
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	backend/src/database/database-health.service.spec.ts Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/backend/src/database/database-health.service.spec.ts

TC-KMS-HEALTH-001 | PASS (CI suite)

Liveness is available without checking PostgreSQL

Requirement: Not recorded | Area: Health endpoints | Level: Unit | Priority: Not recorded | Kind: Automated

Preconditions	Test fixture/mocks as defined by the source file
Input / steps	Execute the named automated assertion in the source file
Expected result	Liveness is available without checking PostgreSQL
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	backend/src/health/health.controller.spec.ts Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/backend/src/health/health.controller.spec.ts

TC-KMS-HEALTH-002 | PASS (CI suite)

Readiness succeeds when PostgreSQL is ready

Requirement: Not recorded | Area: Health endpoints | Level: Unit | Priority: Not recorded | Kind: Automated

Preconditions	Test fixture/mocks as defined by the source file
Input / steps	Execute the named automated assertion in the source file
Expected result	Readiness succeeds when PostgreSQL is ready
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	backend/src/health/health.controller.spec.ts Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/backend/src/health/health.controller.spec.ts

TC-KMS-HEALTH-003 | PASS (CI suite)

Readiness returns service-unavailable semantics when PostgreSQL is unavailable

Requirement: Not recorded | Area: Health endpoints | Level: Unit | Priority: Not recorded | Kind: Automated

Preconditions	Test fixture/mocks as defined by the source file
Input / steps	Execute the named automated assertion in the source file
Expected result	Readiness returns service-unavailable semantics when PostgreSQL is unavailable
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	backend/src/health/health.controller.spec.ts Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/backend/src/health/health.controller.spec.ts

TC-KMS-MENU-001 | PASS (CI suite)

Menu stock enum contains only approved statuses

Requirement: Not recorded | Area: Menu data model / repository | Level: Unit | Priority: Not recorded | Kind: Automated

Preconditions	Test fixture/mocks as defined by the source file
Input / steps	Execute the named automated assertion in the source file
Expected result	Menu stock enum contains only approved statuses
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	backend/src/modules/menu/enums/menu-item-stock-status.enum.spec.ts Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/backend/src/modules/menu/enums/menu-item-stock-status.enum.spec.ts

TC-KMS-MENU-002 | PASS (CI suite)

MealCategory maps to the approved KMS table

Requirement: Not recorded | Area: Menu data model / repository | Level: Unit | Priority: Not recorded | Kind: Automated

Preconditions	Test fixture/mocks as defined by the source file
Input / steps	Execute the named automated assertion in the source file
Expected result	MealCategory maps to the approved KMS table
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	backend/src/modules/menu-categories/entities/meal-category.entity.spec.ts Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/backend/src/modules/menu-categories/entities/meal-category.entity.spec.ts

TC-KMS-MENU-003 | PASS (CI suite)

MealCategory fields map to approved columns

Requirement: Not recorded | Area: Menu data model / repository | Level: Unit | Priority: Not recorded | Kind: Automated

Preconditions	Test fixture/mocks as defined by the source file
Input / steps	Execute the named automated assertion in the source file
Expected result	MealCategory fields map to approved columns
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	backend/src/modules/menu-categories/entities/meal-category.entity.spec.ts Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/backend/src/modules/menu-categories/entities/meal-category.entity.spec.ts

TC-KMS-MENU-004 | PASS (CI suite)

MealCategory primary key uses UUID generation

Requirement: Not recorded | Area: Menu data model / repository | Level: Unit | Priority: Not recorded | Kind: Automated

Preconditions	Test fixture/mocks as defined by the source file
Input / steps	Execute the named automated assertion in the source file
Expected result	MealCategory primary key uses UUID generation
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	backend/src/modules/menu-categories/entities/meal-category.entity.spec.ts Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/backend/src/modules/menu-categories/entities/meal-category.entity.spec.ts

TC-KMS-MENU-005 | PASS (CI suite)

MealCategory has a one-to-many relationship with MenuItems

Requirement: Not recorded | Area: Menu data model / repository | Level: Unit | Priority: Not recorded | Kind: Automated

Preconditions	Test fixture/mocks as defined by the source file
Input / steps	Execute the named automated assertion in the source file
Expected result	MealCategory has a one-to-many relationship with MenuItems
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	backend/src/modules/menu-categories/entities/meal-category.entity.spec.ts Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/backend/src/modules/menu-categories/entities/meal-category.entity.spec.ts

TC-KMS-MENU-006 | PASS (CI suite)

MenuItem maps to the approved KMS table

Requirement: Not recorded | Area: Menu data model / repository | Level: Unit | Priority: Not recorded | Kind: Automated

Preconditions	Test fixture/mocks as defined by the source file
Input / steps	Execute the named automated assertion in the source file
Expected result	MenuItem maps to the approved KMS table
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	backend/src/modules/menu/entities/menu-item.entity.spec.ts Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/backend/src/modules/menu/entities/menu-item.entity.spec.ts

TC-KMS-MENU-007 | PASS (CI suite)

MenuItem core fields have approved mappings

Requirement: Not recorded | Area: Menu data model / repository | Level: Unit | Priority: Not recorded | Kind: Automated

Preconditions	Test fixture/mocks as defined by the source file
Input / steps	Execute the named automated assertion in the source file
Expected result	MenuItem core fields have approved mappings
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	backend/src/modules/menu/entities/menu-item.entity.spec.ts Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/backend/src/modules/menu/entities/menu-item.entity.spec.ts

TC-KMS-MENU-008 | PASS (CI suite)

MenuItem optional columns follow approved nullability

Requirement: Not recorded | Area: Menu data model / repository | Level: Unit | Priority: Not recorded | Kind: Automated

Preconditions	Test fixture/mocks as defined by the source file
Input / steps	Execute the named automated assertion in the source file
Expected result	MenuItem optional columns follow approved nullability
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	backend/src/modules/menu/entities/menu-item.entity.spec.ts Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/backend/src/modules/menu/entities/menu-item.entity.spec.ts

TC-KMS-MENU-009 | PASS (CI suite)

MenuItem maps stock status to approved PostgreSQL enum and default

Requirement: Not recorded | Area: Menu data model / repository | Level: Unit | Priority: Not recorded | Kind: Automated

Preconditions	Test fixture/mocks as defined by the source file
Input / steps	Execute the named automated assertion in the source file
Expected result	MenuItem maps stock status to approved PostgreSQL enum and default
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	backend/src/modules/menu/entities/menu-item.entity.spec.ts Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/backend/src/modules/menu/entities/menu-item.entity.spec.ts

TC-KMS-MENU-010 | PASS (CI suite)

MenuItem category is a required many-to-one relation

Requirement: Not recorded | Area: Menu data model / repository | Level: Unit | Priority: Not recorded | Kind: Automated

Preconditions	Test fixture/mocks as defined by the source file
Input / steps	Execute the named automated assertion in the source file
Expected result	MenuItem category is a required many-to-one relation
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	backend/src/modules/menu/entities/menu-item.entity.spec.ts Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/backend/src/modules/menu/entities/menu-item.entity.spec.ts

TC-KMS-MENU-011 | PASS (CI suite)

MenuItem does not store allergens, dietary labels or calories in denormalized columns

Requirement: Not recorded | Area: Menu data model / repository | Level: Unit | Priority: Not recorded | Kind: Automated

Preconditions	Test fixture/mocks as defined by the source file
Input / steps	Execute the named automated assertion in the source file
Expected result	MenuItem does not store allergens, dietary labels or calories in denormalized columns
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	backend/src/modules/menu/entities/menu-item.entity.spec.ts Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/backend/src/modules/menu/entities/menu-item.entity.spec.ts

TC-KMS-MENU-012 | PASS (CI suite)

Categories are retrieved in alphabetical order

Requirement: Not recorded | Area: Menu data model / repository | Level: Unit | Priority: Not recorded | Kind: Automated

Preconditions	Test fixture/mocks as defined by the source file
Input / steps	Execute the named automated assertion in the source file
Expected result	Categories are retrieved in alphabetical order
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	backend/src/modules/menu-categories/repositories/meal-category.repository.spec.ts Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/backend/src/modules/menu-categories/repositories/meal-category.repository.spec.ts

TC-KMS-MENU-013 | PASS (CI suite)

Category lookup is filtered by identifier

Requirement: Not recorded | Area: Menu data model / repository | Level: Unit | Priority: Not recorded | Kind: Automated

Preconditions	Test fixture/mocks as defined by the source file
Input / steps	Execute the named automated assertion in the source file
Expected result	Category lookup is filtered by identifier
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	backend/src/modules/menu-categories/repositories/meal-category.repository.spec.ts Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/backend/src/modules/menu-categories/repositories/meal-category.repository.spec.ts

TC-KMS-MENU-014 | PASS (CI suite)

Menu items are retrieved in alphabetical order

Requirement: Not recorded | Area: Menu data model / repository | Level: Unit | Priority: Not recorded | Kind: Automated

Preconditions	Test fixture/mocks as defined by the source file
Input / steps	Execute the named automated assertion in the source file
Expected result	Menu items are retrieved in alphabetical order
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	backend/src/modules/menu/repositories/menu-item.repository.spec.ts Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/backend/src/modules/menu/repositories/menu-item.repository.spec.ts

TC-KMS-MENU-015 | PASS (CI suite)

Menu item lookup is filtered by identifier

Requirement: Not recorded | Area: Menu data model / repository | Level: Unit | Priority: Not recorded | Kind: Automated

Preconditions	Test fixture/mocks as defined by the source file
Input / steps	Execute the named automated assertion in the source file
Expected result	Menu item lookup is filtered by identifier
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	backend/src/modules/menu/repositories/menu-item.repository.spec.ts Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/backend/src/modules/menu/repositories/menu-item.repository.spec.ts

TC-KMS-MENU-016 | PASS (CI suite)

MenuCategoriesModule loads

Requirement: Not recorded | Area: Menu data model / repository | Level: Unit | Priority: Not recorded | Kind: Automated

Preconditions	Test fixture/mocks as defined by the source file
Input / steps	Execute the named automated assertion in the source file
Expected result	MenuCategoriesModule loads
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	backend/src/modules/menu-categories/menu-categories.module.spec.ts Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/backend/src/modules/menu-categories/menu-categories.module.spec.ts

TC-KMS-MENU-017 | PASS (CI suite)

MenuModule loads

Requirement: Not recorded | Area: Menu data model / repository | Level: Unit | Priority: Not recorded | Kind: Automated

Preconditions	Test fixture/mocks as defined by the source file
Input / steps	Execute the named automated assertion in the source file
Expected result	MenuModule loads
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	backend/src/modules/menu/menu.module.spec.ts Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/backend/src/modules/menu/menu.module.spec.ts

TC-KMS-MENU-018 | PASS (CI suite)

Allergen entity maps to approved table

Requirement: Not recorded | Area: Menu data model / repository | Level: Unit | Priority: Not recorded | Kind: Automated

Preconditions	Test fixture/mocks as defined by the source file
Input / steps	Execute the named automated assertion in the source file
Expected result	Allergen entity maps to approved table
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	backend/src/modules/menu/entities/allergen.entity.spec.ts Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/backend/src/modules/menu/entities/allergen.entity.spec.ts

TC-KMS-MENU-019 | PASS (CI suite)

Allergen UUID and unique name have approved mappings

Requirement: Not recorded | Area: Menu data model / repository | Level: Unit | Priority: Not recorded | Kind: Automated

Preconditions	Test fixture/mocks as defined by the source file
Input / steps	Execute the named automated assertion in the source file
Expected result	Allergen UUID and unique name have approved mappings
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	backend/src/modules/menu/entities/allergen.entity.spec.ts Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/backend/src/modules/menu/entities/allergen.entity.spec.ts

TC-KMS-MENU-020 | PASS (CI suite)

Allergen has a many-to-many relation to MenuItem

Requirement: Not recorded | Area: Menu data model / repository | Level: Unit | Priority: Not recorded | Kind: Automated

Preconditions	Test fixture/mocks as defined by the source file
Input / steps	Execute the named automated assertion in the source file
Expected result	Allergen has a many-to-many relation to MenuItem
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	backend/src/modules/menu/entities/allergen.entity.spec.ts Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/backend/src/modules/menu/entities/allergen.entity.spec.ts

TC-KMS-MENU-021 | PASS (CI suite)

DietaryLabel entity maps to approved table

Requirement: Not recorded | Area: Menu data model / repository | Level: Unit | Priority: Not recorded | Kind: Automated

Preconditions	Test fixture/mocks as defined by the source file
Input / steps	Execute the named automated assertion in the source file
Expected result	DietaryLabel entity maps to approved table
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	backend/src/modules/menu/entities/dietary-label.entity.spec.ts Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/backend/src/modules/menu/entities/dietary-label.entity.spec.ts

TC-KMS-MENU-022 | PASS (CI suite)

DietaryLabel UUID and unique name have approved mappings

Requirement: Not recorded | Area: Menu data model / repository | Level: Unit | Priority: Not recorded | Kind: Automated

Preconditions	Test fixture/mocks as defined by the source file
Input / steps	Execute the named automated assertion in the source file
Expected result	DietaryLabel UUID and unique name have approved mappings
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	backend/src/modules/menu/entities/dietary-label.entity.spec.ts Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/backend/src/modules/menu/entities/dietary-label.entity.spec.ts

TC-KMS-MENU-023 | PASS (CI suite)

DietaryLabel has a many-to-many relation to MenuItem

Requirement: Not recorded | Area: Menu data model / repository | Level: Unit | Priority: Not recorded | Kind: Automated

Preconditions	Test fixture/mocks as defined by the source file
Input / steps	Execute the named automated assertion in the source file
Expected result	DietaryLabel has a many-to-many relation to MenuItem
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	backend/src/modules/menu/entities/dietary-label.entity.spec.ts Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/backend/src/modules/menu/entities/dietary-label.entity.spec.ts

TC-KMS-MENU-024 | PASS (CI suite)

MenuItem uses the approved allergen junction table

Requirement: Not recorded | Area: Menu data model / repository | Level: Unit | Priority: Not recorded | Kind: Automated

Preconditions	Test fixture/mocks as defined by the source file
Input / steps	Execute the named automated assertion in the source file
Expected result	MenuItem uses the approved allergen junction table
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	backend/src/modules/menu/entities/menu-item.entity.spec.ts Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/backend/src/modules/menu/entities/menu-item.entity.spec.ts

TC-KMS-MENU-025 | PASS (CI suite)

MenuItem uses the approved dietary-label junction table

Requirement: Not recorded | Area: Menu data model / repository | Level: Unit | Priority: Not recorded | Kind: Automated

Preconditions	Test fixture/mocks as defined by the source file
Input / steps	Execute the named automated assertion in the source file
Expected result	MenuItem uses the approved dietary-label junction table
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	backend/src/modules/menu/entities/menu-item.entity.spec.ts Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/backend/src/modules/menu/entities/menu-item.entity.spec.ts

TC-KMS-MENU-026 | PASS (CI suite)

Allergens are retrieved alphabetically

Requirement: Not recorded | Area: Menu data model / repository | Level: Unit | Priority: Not recorded | Kind: Automated

Preconditions	Test fixture/mocks as defined by the source file
Input / steps	Execute the named automated assertion in the source file
Expected result	Allergens are retrieved alphabetically
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	backend/src/modules/menu/repositories/allergen.repository.spec.ts Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/backend/src/modules/menu/repositories/allergen.repository.spec.ts

TC-KMS-MENU-027 | PASS (CI suite)

Allergen lookup is filtered by identifier

Requirement: Not recorded | Area: Menu data model / repository | Level: Unit | Priority: Not recorded | Kind: Automated

Preconditions	Test fixture/mocks as defined by the source file
Input / steps	Execute the named automated assertion in the source file
Expected result	Allergen lookup is filtered by identifier
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	backend/src/modules/menu/repositories/allergen.repository.spec.ts Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/backend/src/modules/menu/repositories/allergen.repository.spec.ts

TC-KMS-MENU-028 | PASS (CI suite)

Dietary labels are retrieved alphabetically

Requirement: Not recorded | Area: Menu data model / repository | Level: Unit | Priority: Not recorded | Kind: Automated

Preconditions	Test fixture/mocks as defined by the source file
Input / steps	Execute the named automated assertion in the source file
Expected result	Dietary labels are retrieved alphabetically
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	backend/src/modules/menu/repositories/dietary-label.repository.spec.ts Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/backend/src/modules/menu/repositories/dietary-label.repository.spec.ts

TC-KMS-MENU-029 | PASS (CI suite)

Dietary-label lookup is filtered by identifier

Requirement: Not recorded | Area: Menu data model / repository | Level: Unit | Priority: Not recorded | Kind: Automated

Preconditions	Test fixture/mocks as defined by the source file
Input / steps	Execute the named automated assertion in the source file
Expected result	Dietary-label lookup is filtered by identifier
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	backend/src/modules/menu/repositories/dietary-label.repository.spec.ts Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/backend/src/modules/menu/repositories/dietary-label.repository.spec.ts

TC-KMS-ORDER-001 | PASS (CI suite)

Food order status enum contains only approved statuses

Requirement: Not recorded | Area: Order data model / repository | Level: Unit | Priority: Not recorded | Kind: Automated

Preconditions	Test fixture/mocks as defined by the source file
Input / steps	Execute the named automated assertion in the source file
Expected result	Food order status enum contains only approved statuses
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	backend/src/modules/orders/enums/food-order-status.enum.spec.ts Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/backend/src/modules/orders/enums/food-order-status.enum.spec.ts

TC-KMS-ORDER-002 | PASS (CI suite)

Meal-period enum contains only approved periods

Requirement: Not recorded | Area: Order data model / repository | Level: Unit | Priority: Not recorded | Kind: Automated

Preconditions	Test fixture/mocks as defined by the source file
Input / steps	Execute the named automated assertion in the source file
Expected result	Meal-period enum contains only approved periods
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	backend/src/modules/orders/enums/food-order-meal-period.enum.spec.ts Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/backend/src/modules/orders/enums/food-order-meal-period.enum.spec.ts

TC-KMS-ORDER-003 | PASS (CI suite)

FoodOrder maps to kms_food_orders

Requirement: Not recorded | Area: Order data model / repository | Level: Unit | Priority: Not recorded | Kind: Automated

Preconditions	Test fixture/mocks as defined by the source file
Input / steps	Execute the named automated assertion in the source file
Expected result	FoodOrder maps to kms_food_orders
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	backend/src/modules/orders/entities/food-order.entity.spec.ts Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/backend/src/modules/orders/entities/food-order.entity.spec.ts

TC-KMS-ORDER-004 | PASS (CI suite)

FoodOrder identifiers and kitchen-display fields use approved columns

Requirement: Not recorded | Area: Order data model / repository | Level: Unit | Priority: Not recorded | Kind: Automated

Preconditions	Test fixture/mocks as defined by the source file
Input / steps	Execute the named automated assertion in the source file
Expected result	FoodOrder identifiers and kitchen-display fields use approved columns
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	backend/src/modules/orders/entities/food-order.entity.spec.ts Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/backend/src/modules/orders/entities/food-order.entity.spec.ts

TC-KMS-ORDER-005 | PASS (CI suite)

FoodOrder status and meal-period PostgreSQL enums and defaults are approved

Requirement: Not recorded | Area: Order data model / repository | Level: Unit | Priority: Not recorded | Kind: Automated

Preconditions	Test fixture/mocks as defined by the source file
Input / steps	Execute the named automated assertion in the source file
Expected result	FoodOrder status and meal-period PostgreSQL enums and defaults are approved
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	backend/src/modules/orders/entities/food-order.entity.spec.ts Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/backend/src/modules/orders/entities/food-order.entity.spec.ts

TC-KMS-ORDER-006 | PASS (CI suite)

FoodOrder monetary fields and nullability are mapped correctly

Requirement: Not recorded | Area: Order data model / repository | Level: Unit | Priority: Not recorded | Kind: Automated

Preconditions	Test fixture/mocks as defined by the source file
Input / steps	Execute the named automated assertion in the source file
Expected result	FoodOrder monetary fields and nullability are mapped correctly
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	backend/src/modules/orders/entities/food-order.entity.spec.ts Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/backend/src/modules/orders/entities/food-order.entity.spec.ts

TC-KMS-ORDER-007 | PASS (CI suite)

FoodOrder total check prohibits negative totals

Requirement: Not recorded | Area: Order data model / repository | Level: Unit | Priority: Not recorded | Kind: Automated

Preconditions	Test fixture/mocks as defined by the source file
Input / steps	Execute the named automated assertion in the source file
Expected result	FoodOrder total check prohibits negative totals
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	backend/src/modules/orders/entities/food-order.entity.spec.ts Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/backend/src/modules/orders/entities/food-order.entity.spec.ts

TC-KMS-ORDER-008 | PASS (CI suite)

FoodOrder exposes order items through a one-to-many relation

Requirement: Not recorded | Area: Order data model / repository | Level: Unit | Priority: Not recorded | Kind: Automated

Preconditions	Test fixture/mocks as defined by the source file
Input / steps	Execute the named automated assertion in the source file
Expected result	FoodOrder exposes order items through a one-to-many relation
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	backend/src/modules/orders/entities/food-order.entity.spec.ts Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/backend/src/modules/orders/entities/food-order.entity.spec.ts

TC-KMS-ORDER-009 | PASS (CI suite)

FoodOrder does not store order items as JSON/JSONB

Requirement: Not recorded | Area: Order data model / repository | Level: Unit | Priority: Not recorded | Kind: Automated

Preconditions	Test fixture/mocks as defined by the source file
Input / steps	Execute the named automated assertion in the source file
Expected result	FoodOrder does not store order items as JSON/JSONB
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	backend/src/modules/orders/entities/food-order.entity.spec.ts Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/backend/src/modules/orders/entities/food-order.entity.spec.ts

TC-KMS-ORDER-010 | PASS (CI suite)

FoodOrderItem maps to kms_food_order_items

Requirement: Not recorded | Area: Order data model / repository | Level: Unit | Priority: Not recorded | Kind: Automated

Preconditions	Test fixture/mocks as defined by the source file
Input / steps	Execute the named automated assertion in the source file
Expected result	FoodOrderItem maps to kms_food_order_items
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	backend/src/modules/orders/entities/food-order-item.entity.spec.ts Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/backend/src/modules/orders/entities/food-order-item.entity.spec.ts

TC-KMS-ORDER-011 | PASS (CI suite)

FoodOrderItem identifiers use canonical UUID columns

Requirement: Not recorded | Area: Order data model / repository | Level: Unit | Priority: Not recorded | Kind: Automated

Preconditions	Test fixture/mocks as defined by the source file
Input / steps	Execute the named automated assertion in the source file
Expected result	FoodOrderItem identifiers use canonical UUID columns
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	backend/src/modules/orders/entities/food-order-item.entity.spec.ts Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/backend/src/modules/orders/entities/food-order-item.entity.spec.ts

TC-KMS-ORDER-012 | PASS (CI suite)

FoodOrderItem persists historical name and unit-price snapshots

Requirement: Not recorded | Area: Order data model / repository | Level: Unit | Priority: Not recorded | Kind: Automated

Preconditions	Test fixture/mocks as defined by the source file
Input / steps	Execute the named automated assertion in the source file
Expected result	FoodOrderItem persists historical name and unit-price snapshots
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	backend/src/modules/orders/entities/food-order-item.entity.spec.ts Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/backend/src/modules/orders/entities/food-order-item.entity.spec.ts

TC-KMS-ORDER-013 | PASS (CI suite)

FoodOrderItem quantity and note use canonical mappings

Requirement: Not recorded | Area: Order data model / repository | Level: Unit | Priority: Not recorded | Kind: Automated

Preconditions	Test fixture/mocks as defined by the source file
Input / steps	Execute the named automated assertion in the source file
Expected result	FoodOrderItem quantity and note use canonical mappings
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	backend/src/modules/orders/entities/food-order-item.entity.spec.ts Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/backend/src/modules/orders/entities/food-order-item.entity.spec.ts

TC-KMS-ORDER-014 | PASS (CI suite)

FoodOrderItem quantity and unit price require positive values

Requirement: Not recorded | Area: Order data model / repository | Level: Unit | Priority: Not recorded | Kind: Automated

Preconditions	Test fixture/mocks as defined by the source file
Input / steps	Execute the named automated assertion in the source file
Expected result	FoodOrderItem quantity and unit price require positive values
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	backend/src/modules/orders/entities/food-order-item.entity.spec.ts Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/backend/src/modules/orders/entities/food-order-item.entity.spec.ts

TC-KMS-ORDER-015 | PASS (CI suite)

FoodOrderItem required order relation cascades on FoodOrder deletion

Requirement: Not recorded | Area: Order data model / repository | Level: Unit | Priority: Not recorded | Kind: Automated

Preconditions	Test fixture/mocks as defined by the source file
Input / steps	Execute the named automated assertion in the source file
Expected result	FoodOrderItem required order relation cascades on FoodOrder deletion
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	backend/src/modules/orders/entities/food-order-item.entity.spec.ts Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/backend/src/modules/orders/entities/food-order-item.entity.spec.ts

TC-KMS-ORDER-016 | PASS (CI suite)

FoodOrderItem references MenuItem

Requirement: Not recorded | Area: Order data model / repository | Level: Unit | Priority: Not recorded | Kind: Automated

Preconditions	Test fixture/mocks as defined by the source file
Input / steps	Execute the named automated assertion in the source file
Expected result	FoodOrderItem references MenuItem
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	backend/src/modules/orders/entities/food-order-item.entity.spec.ts Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/backend/src/modules/orders/entities/food-order-item.entity.spec.ts

TC-KMS-ORDER-017 | PASS (CI suite)

FoodOrderItem does not store JSON/JSONB

Requirement: Not recorded | Area: Order data model / repository | Level: Unit | Priority: Not recorded | Kind: Automated

Preconditions	Test fixture/mocks as defined by the source file
Input / steps	Execute the named automated assertion in the source file
Expected result	FoodOrderItem does not store JSON/JSONB
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	backend/src/modules/orders/entities/food-order-item.entity.spec.ts Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/backend/src/modules/orders/entities/food-order-item.entity.spec.ts

TC-KMS-ORDER-018 | PASS (CI suite)

Orders are retrieved by placement time

Requirement: Not recorded | Area: Order data model / repository | Level: Unit | Priority: Not recorded | Kind: Automated

Preconditions	Test fixture/mocks as defined by the source file
Input / steps	Execute the named automated assertion in the source file
Expected result	Orders are retrieved by placement time
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	backend/src/modules/orders/repositories/food-order.repository.spec.ts Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/backend/src/modules/orders/repositories/food-order.repository.spec.ts

TC-KMS-ORDER-019 | PASS (CI suite)

Order lookup is filtered by identifier

Requirement: Not recorded | Area: Order data model / repository | Level: Unit | Priority: Not recorded | Kind: Automated

Preconditions	Test fixture/mocks as defined by the source file
Input / steps	Execute the named automated assertion in the source file
Expected result	Order lookup is filtered by identifier
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	backend/src/modules/orders/repositories/food-order.repository.spec.ts Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/backend/src/modules/orders/repositories/food-order.repository.spec.ts

TC-KMS-ORDER-020 | PASS (CI suite)

Order-item lookup is filtered by identifier

Requirement: Not recorded | Area: Order data model / repository | Level: Unit | Priority: Not recorded | Kind: Automated

Preconditions	Test fixture/mocks as defined by the source file
Input / steps	Execute the named automated assertion in the source file
Expected result	Order-item lookup is filtered by identifier
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	backend/src/modules/orders/repositories/food-order-item.repository.spec.ts Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/backend/src/modules/orders/repositories/food-order-item.repository.spec.ts

TC-KMS-ORDER-021 | PASS (CI suite)

Order line items are retrieved by order ID

Requirement: Not recorded | Area: Order data model / repository | Level: Unit | Priority: Not recorded | Kind: Automated

Preconditions	Test fixture/mocks as defined by the source file
Input / steps	Execute the named automated assertion in the source file
Expected result	Order line items are retrieved by order ID
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	backend/src/modules/orders/repositories/food-order-item.repository.spec.ts Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/backend/src/modules/orders/repositories/food-order-item.repository.spec.ts

TC-KMS-ORDER-022 | PASS (CI suite)

OrdersModule loads

Requirement: Not recorded | Area: Order data model / repository | Level: Unit | Priority: Not recorded | Kind: Automated

Preconditions	Test fixture/mocks as defined by the source file
Input / steps	Execute the named automated assertion in the source file
Expected result	OrdersModule loads
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	backend/src/modules/orders/orders.module.spec.ts Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/backend/src/modules/orders/orders.module.spec.ts

TC-KMS-APP-E2E-001 | PASS (CI suite)

GET / responds successfully with scaffold greeting

Requirement: Not recorded | Area: HTTP & health | Level: E2E (application) | Priority: Not recorded | Kind: Automated

Preconditions	Nest application initialized with isolated PostgreSQL test service
Input / steps	Execute the named automated assertion in the source file
Expected result	GET / responds successfully with scaffold greeting
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	backend/test/app.e2e-spec.ts Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/backend/test/app.e2e-spec.ts

TC-KMS-APP-E2E-002 | PASS (CI suite)

GET /health/live responds with status ok

Requirement: Not recorded | Area: HTTP & health | Level: E2E (application) | Priority: Not recorded | Kind: Automated

Preconditions	Nest application initialized with isolated PostgreSQL test service
Input / steps	Execute the named automated assertion in the source file
Expected result	GET /health/live responds with status ok
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	backend/test/app.e2e-spec.ts Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/backend/test/app.e2e-spec.ts

TC-KMS-APP-E2E-003 | PASS (CI suite)

GET /health/ready responds with database up

Requirement: Not recorded | Area: HTTP & health | Level: E2E (application) | Priority: Not recorded | Kind: Automated

Preconditions	Nest application initialized with isolated PostgreSQL test service
Input / steps	Execute the named automated assertion in the source file
Expected result	GET /health/ready responds with database up
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	backend/test/app.e2e-spec.ts Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/backend/test/app.e2e-spec.ts

TC-KMS-DB-E2E-001 | PASS (CI suite)

TypeORM data source initializes with PostgreSQL

Requirement: Not recorded | Area: Database E2E | Level: E2E (PostgreSQL) | Priority: Not recorded | Kind: Automated

Preconditions	GitHub Actions PostgreSQL 15 ephemeral service
Input / steps	Execute the named automated assertion in the source file
Expected result	TypeORM data source initializes with PostgreSQL
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	backend/test/database.e2e-spec.ts Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/backend/test/database.e2e-spec.ts

TC-KMS-DB-E2E-002 | PASS (CI suite)

PostgreSQL connectivity query succeeds

Requirement: Not recorded | Area: Database E2E | Level: E2E (PostgreSQL) | Priority: Not recorded | Kind: Automated

Preconditions	GitHub Actions PostgreSQL 15 ephemeral service
Input / steps	Execute the named automated assertion in the source file
Expected result	PostgreSQL connectivity query succeeds
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	backend/test/database.e2e-spec.ts Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/backend/test/database.e2e-spec.ts

TC-KMS-FE-001 | PASS (CI suite)

Foundation page renders the KMS heading

Requirement: Not recorded | Area: Frontend foundation | Level: Frontend component | Priority: Not recorded | Kind: Automated

Preconditions	Jest/JSDOM React Testing Library test environment
Input / steps	Execute the named automated assertion in the source file
Expected result	Foundation page renders the KMS heading
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	frontend/src/app/page.test.tsx Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/frontend/src/app/page.test.tsx

TC-KMS-FE-002 | PASS (CI suite)

Ant Design control can render inside AppProviders

Requirement: Not recorded | Area: Frontend foundation | Level: Frontend component | Priority: Not recorded | Kind: Automated

Preconditions	Jest/JSDOM React Testing Library test environment
Input / steps	Execute the named automated assertion in the source file
Expected result	Ant Design control can render inside AppProviders
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	frontend/src/providers/app-providers.test.tsx Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/frontend/src/providers/app-providers.test.tsx

TC-KMS-FE-003 | PASS (CI suite)

Shared theme exposes approved tokens and accessible base control height

Requirement: Not recorded | Area: Frontend foundation | Level: Frontend component | Priority: Not recorded | Kind: Automated

Preconditions	Jest/JSDOM React Testing Library test environment
Input / steps	Execute the named automated assertion in the source file
Expected result	Shared theme exposes approved tokens and accessible base control height
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	frontend/src/providers/app-providers.test.tsx Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/frontend/src/providers/app-providers.test.tsx

TC-KMS-FE-004 | PASS (CI suite)

Shell exposes the KMS heading and main landmark

Requirement: Not recorded | Area: Frontend foundation | Level: Frontend component | Priority: Not recorded | Kind: Automated

Preconditions	Jest/JSDOM React Testing Library test environment
Input / steps	Execute the named automated assertion in the source file
Expected result	Shell exposes the KMS heading and main landmark
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	frontend/src/components/layout/kms-shell.test.tsx Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/frontend/src/components/layout/kms-shell.test.tsx

TC-KMS-FE-005 | PASS (CI suite)

Keyboard-accessible skip link focuses main content

Requirement: Not recorded | Area: Frontend foundation | Level: Frontend component | Priority: Not recorded | Kind: Automated

Preconditions	Jest/JSDOM React Testing Library test environment
Input / steps	Execute the named automated assertion in the source file
Expected result	Keyboard-accessible skip link focuses main content
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	frontend/src/components/layout/kms-shell.test.tsx Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/frontend/src/components/layout/kms-shell.test.tsx

TC-KMS-FE-006 | PASS (CI suite)

Shell renders the kitchen operations footer

Requirement: Not recorded | Area: Frontend foundation | Level: Frontend component | Priority: Not recorded | Kind: Automated

Preconditions	Jest/JSDOM React Testing Library test environment
Input / steps	Execute the named automated assertion in the source file
Expected result	Shell renders the kitchen operations footer
Recorded actual result	The containing test suite passed in the evaluated GitHub Actions run.
Test file / evidence	frontend/src/components/layout/kms-shell.test.tsx Evidence reference https://github.com/forever-hotel/kitchen-management/blob/89087002ac2df8f07b6140c00111979c75bc2799/frontend/src/components/layout/kms-shell.test.tsx

KMS-ACC-01 | NOT TESTED

Place new guest food order through live KMS UI and update status

Requirement: Not recorded | Area: Unverified acceptance | Level: Cross-service / acceptance | Priority: Not recorded | Kind: Acceptance gap

Preconditions	Staging with services and safe test data
Input / steps	Execute an agreed acceptance scenario with observable outcome
Expected result	Place new guest food order through live KMS UI and update status
Recorded actual result	No completed workflow/UI test in evaluated develop snapshot
Test file / evidence	(No verified automated test)
Additional note	On develop 1f56edd2ff (8 Nov 2026, Run 42502180570), KMS expanded to 665 passing subsystem unit/E2E/frontend tests (including orders-create, orders-read, orders-transition, allergy-safety, ordering-window, and menu CRUD E2E suites); live cross-service staging and physical KOT hardware remain unverified.

KMS-ACC-02 | NOT TESTED

Accept FOSS order event via actual inter-service messaging

Requirement: Not recorded | Area: Unverified acceptance | Level: Cross-service / acceptance | Priority: Not recorded | Kind: Acceptance gap

Preconditions	Staging with services and safe test data
Input / steps	Execute an agreed acceptance scenario with observable outcome
Expected result	Accept FOSS order event via actual inter-service messaging
Recorded actual result	Messaging directory is only a placeholder on evaluated develop
Test file / evidence	(No verified automated test)
Additional note	On develop 1f56edd2ff (8 Nov 2026, Run 42502180570), KMS expanded to 665 passing subsystem unit/E2E/frontend tests (including orders-create, orders-read, orders-transition, allergy-safety, ordering-window, and menu CRUD E2E suites); live cross-service staging and physical KOT hardware remain unverified.

KMS-ACC-03 | NOT TESTED

Print a real KOT and verify printer/persistence/retry

Requirement: Not recorded | Area: Unverified acceptance | Level: Cross-service / acceptance | Priority: Not recorded | Kind: Acceptance gap

Preconditions	Staging with services and safe test data
Input / steps	Execute an agreed acceptance scenario with observable outcome
Expected result	Print a real KOT and verify printer/persistence/retry
Recorded actual result	KOT printing acceptance is not evidenced on evaluated develop
Test file / evidence	(No verified automated test)
Additional note	On develop 1f56edd2ff (8 Nov 2026, Run 42502180570), KMS expanded to 665 passing subsystem unit/E2E/frontend tests (including orders-create, orders-read, orders-transition, allergy-safety, ordering-window, and menu CRUD E2E suites); live cross-service staging and physical KOT hardware remain unverified.

KMS-ACC-04 | NOT TESTED

Dispatch completed food order to WKMS and verify delivery

Requirement: Not recorded | Area: Unverified acceptance | Level: Cross-service / acceptance | Priority: Not recorded | Kind: Acceptance gap

Preconditions	Staging with services and safe test data
Input / steps	Execute an agreed acceptance scenario with observable outcome
Expected result	Dispatch completed food order to WKMS and verify delivery
Recorded actual result	Worker dispatch integration not evidenced on evaluated develop
Test file / evidence	(No verified automated test)
Additional note	On develop 1f56edd2ff (8 Nov 2026, Run 42502180570), KMS expanded to 665 passing subsystem unit/E2E/frontend tests (including orders-create, orders-read, orders-transition, allergy-safety, ordering-window, and menu CRUD E2E suites); live cross-service staging and physical KOT hardware remain unverified.

KMS-ACC-05 | NOT TESTED

Run approved concurrency and latency tests against a staged KMS

Requirement: Not recorded | Area: Unverified acceptance | Level: Cross-service / acceptance | Priority: Not recorded | Kind: Acceptance gap

Preconditions	Staging with services and safe test data
Input / steps	Execute an agreed acceptance scenario with observable outcome
Expected result	Run approved concurrency and latency tests against a staged KMS
Recorded actual result	No timed load results or harness in evaluated develop
Test file / evidence	(No verified automated test)
Additional note	On develop 1f56edd2ff (8 Nov 2026, Run 42502180570), KMS expanded to 665 passing subsystem unit/E2E/frontend tests (including orders-create, orders-read, orders-transition, allergy-safety, ordering-window, and menu CRUD E2E suites); live cross-service staging and physical KOT hardware remain unverified.

KMS-ACC-06 | NOT TESTED

Prove authenticated role-specific access to operational endpoints

Requirement: Not recorded | Area: Unverified acceptance | Level: Cross-service / acceptance | Priority: Not recorded | Kind: Acceptance gap

Preconditions	Staging with services and safe test data
Input / steps	Execute an agreed acceptance scenario with observable outcome
Expected result	Prove authenticated role-specific access to operational endpoints
Recorded actual result	No audited KMS authorization tests for operational endpoints in develop
Test file / evidence	(No verified automated test)
Additional note	On develop 1f56edd2ff (8 Nov 2026, Run 42502180570), KMS expanded to 665 passing subsystem unit/E2E/frontend tests (including orders-create, orders-read, orders-transition, allergy-safety, ordering-window, and menu CRUD E2E suites); live cross-service staging and physical KOT hardware remain unverified.

7. Worker Management System (WKMS)

Repository: forever-hotel/worker-management | Branch: develop | Tested revision: 1a33059c0005752ca709ca91ab11067dc6cff55a (Oct 25 register baseline: 0f91f1c9d6390c80c9f12f82b834fee9bf374104) | CI date: 8 November 2026

Evidence: https://github.com/forever-hotel/worker-management/actions/runs/42504698971, https://github.com/forever-hotel/worker-management/actions/runs/42504970394, https://github.com/forever-hotel/worker-management/actions/runs/40174879698

Register entries: 37; CI-suite-linked pass: 30; not tested/not verified: 7. By 8 November 2026 (commit 1a33059c00, Run 42504698971), WKMS automated CI expanded from 445 to 649 passing tests (400 backend unit/integration with 43 skipped, 249 frontend Jest) with >=80% global branch and line coverage enforced in both backend and frontend.

TC-WKMS-001 | PASS (CI suite)

should authenticate an active WORKER and issue a token

Requirement: NFR-12 | Level: Unit | Priority: P1

Preconditions	Prepared isolated fixtures
Input / steps	Run named test with fixture data
Expected result	Named assertions succeed
Recorded actual result	Test suite passed in matched GitHub Actions CI; individual assertion transcript not archived.
Test file / evidence	backend/src/auth/auth.service.spec.ts

TC-WKMS-002 | PASS (CI suite)

should reject a non-WORKER role

Requirement: NFR-12 | Level: Unit | Priority: P1

Preconditions	Prepared isolated fixtures
Input / steps	Run named test with fixture data
Expected result	Named assertions succeed
Recorded actual result	Test suite passed in matched GitHub Actions CI; individual assertion transcript not archived.
Test file / evidence	backend/src/auth/auth.service.spec.ts

TC-WKMS-003 | PASS (CI suite)

rejects expired or invalid JWTs

Requirement: NFR-12 | Level: Unit | Priority: P1

Preconditions	Prepared isolated fixtures
Input / steps	Run named test with fixture data
Expected result	Named assertions succeed
Recorded actual result	Test suite passed in matched GitHub Actions CI; individual assertion transcript not archived.
Test file / evidence	backend/src/realtime/wkms-realtime.gateway.spec.ts

TC-WKMS-004 | PASS (CI suite)

should return available tasks ordered by submission time

Requirement: WM-01 | Level: Unit | Priority: P1

Preconditions	Prepared isolated fixtures
Input / steps	Run named test with fixture data
Expected result	Named assertions succeed
Recorded actual result	Test suite passed in matched GitHub Actions CI; individual assertion transcript not archived.
Test file / evidence	backend/src/database/repositories/task.repository.spec.ts

TC-WKMS-005 | PASS (CI suite)

should return the task queue from the repository

Requirement: WM-01 | Level: Unit | Priority: P1

Preconditions	Prepared isolated fixtures
Input / steps	Run named test with fixture data
Expected result	Named assertions succeed
Recorded actual result	Test suite passed in matched GitHub Actions CI; individual assertion transcript not archived.
Test file / evidence	backend/src/tasks/task.service.spec.ts

TC-WKMS-006 | PASS (CI suite)

renders shared queue data without fetching on mount

Requirement: WM-01 | Level: Frontend Jest | Priority: P1

Preconditions	Prepared isolated fixtures
Input / steps	Run named test with fixture data
Expected result	Named assertions succeed
Recorded actual result	Test suite passed in matched GitHub Actions CI; individual assertion transcript not archived.
Test file / evidence	frontend/src/features/task-queue/components/task-queue-screen.test.tsx

TC-WKMS-007 | PASS (CI suite)

configures an authenticated Socket.IO connection

Requirement: WM-01b | Level: Frontend Jest | Priority: P1

Preconditions	Prepared isolated fixtures
Input / steps	Run named test with fixture data
Expected result	Named assertions succeed
Recorded actual result	Test suite passed in matched GitHub Actions CI; individual assertion transcript not archived.
Test file / evidence	frontend/src/lib/task-socket.test.ts
Additional note	Proves client config; cross-deployed event delivery requires separate test.

TC-WKMS-008 | PASS (CI suite)

registers only the expected realtime event names

Requirement: WM-01b | Level: Frontend Jest | Priority: P1

Preconditions	Prepared isolated fixtures
Input / steps	Run named test with fixture data
Expected result	Named assertions succeed
Recorded actual result	Test suite passed in matched GitHub Actions CI; individual assertion transcript not archived.
Test file / evidence	frontend/src/lib/task-socket.test.ts

TC-WKMS-009 | PASS (CI suite)

loads and renders a real unassigned task

Requirement: WM-02 | Level: Frontend Jest | Priority: P1

Preconditions	Prepared isolated fixtures
Input / steps	Run named test with fixture data
Expected result	Named assertions succeed
Recorded actual result	Test suite passed in matched GitHub Actions CI; individual assertion transcript not archived.
Test file / evidence	frontend/src/features/tasks/components/task-detail-screen.test.tsx

TC-WKMS-010 | NOT TESTED

should reject task detail assigned to another worker

Requirement: WM-02 | Level: Jest local integration | Priority: P1

Preconditions	Prepared isolated fixtures
Input / steps	Run named test with fixture data
Expected result	Named assertions succeed
Recorded actual result	No verified complete execution or acceptance evidence at this commit.
Test file / evidence	backend/test/integration/task.controller.integration.spec.ts
Additional note	Integration suite is not confirmed as executed in the 322-passed CI unit-group; manual follow-up.

TC-WKMS-011 | PASS (CI suite)

should claim an unassigned task for an active worker

Requirement: WM-03 | Level: Unit | Priority: P1

Preconditions	Prepared isolated fixtures
Input / steps	Run named test with fixture data
Expected result	Named assertions succeed
Recorded actual result	Test suite passed in matched GitHub Actions CI; individual assertion transcript not archived.
Test file / evidence	backend/src/database/repositories/task.repository.spec.ts

TC-WKMS-012 | PASS (CI suite)

should reject a claim when worker already has three active tasks

Requirement: WM-03 | Level: Unit | Priority: P1

Preconditions	Prepared isolated fixtures
Input / steps	Run named test with fixture data
Expected result	Named assertions succeed
Recorded actual result	Test suite passed in matched GitHub Actions CI; individual assertion transcript not archived.
Test file / evidence	backend/src/database/repositories/task.repository.spec.ts

TC-WKMS-013 | PASS (CI suite)

should reject a claim when task is not unassigned

Requirement: WM-03 | Level: Unit | Priority: P1

Preconditions	Prepared isolated fixtures
Input / steps	Run named test with fixture data
Expected result	Named assertions succeed
Recorded actual result	Test suite passed in matched GitHub Actions CI; individual assertion transcript not archived.
Test file / evidence	backend/src/database/repositories/task.repository.spec.ts

TC-WKMS-014 | PASS (CI suite)

starts an assigned task

Requirement: WM-04 | Level: Frontend Jest | Priority: P1

Preconditions	Prepared isolated fixtures
Input / steps	Run named test with fixture data
Expected result	Named assertions succeed
Recorded actual result	Test suite passed in matched GitHub Actions CI; individual assertion transcript not archived.
Test file / evidence	frontend/src/features/tasks/components/task-detail-screen.test.tsx

TC-WKMS-015 | PASS (CI suite)

should reject starting a task owned by another worker

Requirement: WM-04 | Level: Unit | Priority: P1

Preconditions	Prepared isolated fixtures
Input / steps	Run named test with fixture data
Expected result	Named assertions succeed
Recorded actual result	Test suite passed in matched GitHub Actions CI; individual assertion transcript not archived.
Test file / evidence	backend/src/database/repositories/task.repository.spec.ts

TC-WKMS-016 | PASS (CI suite)

completes an in-progress task

Requirement: WM-05 | Level: Frontend Jest | Priority: P1

Preconditions	Prepared isolated fixtures
Input / steps	Run named test with fixture data
Expected result	Named assertions succeed
Recorded actual result	Test suite passed in matched GitHub Actions CI; individual assertion transcript not archived.
Test file / evidence	frontend/src/features/tasks/components/task-detail-screen.test.tsx

TC-WKMS-017 | PASS (CI suite)

should reject completing a task owned by another worker

Requirement: WM-05 | Level: Unit | Priority: P1

Preconditions	Prepared isolated fixtures
Input / steps	Run named test with fixture data
Expected result	Named assertions succeed
Recorded actual result	Test suite passed in matched GitHub Actions CI; individual assertion transcript not archived.
Test file / evidence	backend/src/database/repositories/task.repository.spec.ts

TC-WKMS-018 | PASS (CI suite)

renders shared authenticated tasks without fetching on mount

Requirement: WM-06 | Level: Frontend Jest | Priority: P1

Preconditions	Prepared isolated fixtures
Input / steps	Run named test with fixture data
Expected result	Named assertions succeed
Recorded actual result	Test suite passed in matched GitHub Actions CI; individual assertion transcript not archived.
Test file / evidence	frontend/src/features/assignments/components/my-tasks-screen.test.tsx

TC-WKMS-019 | PASS (CI suite)

displays completed tasks from the authenticated shift summary

Requirement: WM-06 | Level: Frontend Jest | Priority: P1

Preconditions	Prepared isolated fixtures
Input / steps	Run named test with fixture data
Expected result	Named assertions succeed
Recorded actual result	Test suite passed in matched GitHub Actions CI; individual assertion transcript not archived.
Test file / evidence	frontend/src/features/assignments/components/my-tasks-screen.test.tsx

TC-WKMS-020 | PASS (CI suite)

broadcasts after successful database escalation

Requirement: WM-07 | Level: Unit | Priority: P1

Preconditions	Prepared isolated fixtures
Input / steps	Run named test with fixture data
Expected result	Named assertions succeed
Recorded actual result	Test suite passed in matched GitHub Actions CI; individual assertion transcript not archived.
Test file / evidence	backend/src/tasks/task-escalation.service.spec.ts

TC-WKMS-021 | PASS (CI suite)

remains safe across repeated scheduler runs

Requirement: WM-07 | Level: Unit | Priority: P1

Preconditions	Prepared isolated fixtures
Input / steps	Run named test with fixture data
Expected result	Named assertions succeed
Recorded actual result	Test suite passed in matched GitHub Actions CI; individual assertion transcript not archived.
Test file / evidence	backend/src/tasks/task-escalation.service.spec.ts

TC-WKMS-022 | PASS (CI suite)

publishes a pending event and marks it published

Requirement: WM-07 | Level: Unit / mocked broker | Priority: P1

Preconditions	Prepared isolated fixtures
Input / steps	Run named test with fixture data
Expected result	Named assertions succeed
Recorded actual result	Test suite passed in matched GitHub Actions CI; individual assertion transcript not archived.
Test file / evidence	backend/src/messaging/outbox/outbox-dispatcher.service.spec.ts
Additional note	Broker/outbox logic unit tested; deployed subscriber reception unverified.

TC-WKMS-023 | NOT TESTED

Worker escalation reflected on FDS UI through real event transport

Requirement: WM-07b | Level: Cross-service E2E | Priority: P1

Preconditions	FDS, WKMS and RabbitMQ in integrated staging
Input / steps	Leave real task unclaimed beyond 15 minutes
Expected result	Escalation appears on FDS within required latency
Recorded actual result	No verified complete execution or acceptance evidence at this commit.
Test file / evidence	backend/src/messaging/publishers/task-escalated.publisher.ts

TC-WKMS-024 | PASS (CI suite)

renders the real worker summary

Requirement: WM-09 | Level: Frontend Jest | Priority: P1

Preconditions	Prepared isolated fixtures
Input / steps	Run named test with fixture data
Expected result	Named assertions succeed
Recorded actual result	Test suite passed in matched GitHub Actions CI; individual assertion transcript not archived.
Test file / evidence	frontend/src/features/shifts/components/shift-dashboard-screen.test.tsx

TC-WKMS-025 | PASS (CI suite)

renders completed task history using real turnaround values

Requirement: WM-09 | Level: Frontend Jest | Priority: P1

Preconditions	Prepared isolated fixtures
Input / steps	Run named test with fixture data
Expected result	Named assertions succeed
Recorded actual result	Test suite passed in matched GitHub Actions CI; individual assertion transcript not archived.
Test file / evidence	frontend/src/features/shifts/components/shift-dashboard-screen.test.tsx

TC-WKMS-026 | PASS (CI suite)

creates a cleaning task from a valid FDS event

Requirement: WM-11 | Level: Unit / mocked DB | Priority: P1

Preconditions	Prepared isolated fixtures
Input / steps	Run named test with fixture data
Expected result	Named assertions succeed
Recorded actual result	Test suite passed in matched GitHub Actions CI; individual assertion transcript not archived.
Test file / evidence	backend/src/messaging/handlers/checkout-completed.handler.spec.ts
Additional note	Does not prove live FDS publisher-to-WKMS consumer E2E.

TC-WKMS-027 | PASS (CI suite)

treats an existing cleaning task as successfully processed

Requirement: WM-11 | Level: Unit | Priority: P1

Preconditions	Prepared isolated fixtures
Input / steps	Run named test with fixture data
Expected result	Named assertions succeed
Recorded actual result	Test suite passed in matched GitHub Actions CI; individual assertion transcript not archived.
Test file / evidence	backend/src/messaging/handlers/checkout-completed.handler.spec.ts

TC-WKMS-028 | PASS (CI suite)

creates a food-delivery task for a valid KMS event

Requirement: WM-01c | Level: Unit / mocked DB | Priority: P1

Preconditions	Prepared isolated fixtures
Input / steps	Run named test with fixture data
Expected result	Named assertions succeed
Recorded actual result	Test suite passed in matched GitHub Actions CI; individual assertion transcript not archived.
Test file / evidence	backend/src/messaging/handlers/order-ready.handler.spec.ts

TC-WKMS-029 | PASS (CI suite)

returns EXISTS when the task was already created

Requirement: WM-01c | Level: Unit | Priority: P1

Preconditions	Prepared isolated fixtures
Input / steps	Run named test with fixture data
Expected result	Named assertions succeed
Recorded actual result	Test suite passed in matched GitHub Actions CI; individual assertion transcript not archived.
Test file / evidence	backend/src/messaging/handlers/order-ready.handler.spec.ts

TC-WKMS-030 | NOT TESTED

Live KMS food-ready event creates worker delivery task

Requirement: WM-01c | Level: Cross-service E2E | Priority: P1

Preconditions	KMS/WKMS integrated deploy and broker
Input / steps	Publish a real order-ready event from KMS
Expected result	WKMS task created and visible to workers
Recorded actual result	No verified complete execution or acceptance evidence at this commit.
Test file / evidence	backend/src/messaging/consumers/order-ready.consumer.ts

TC-WKMS-031 | PASS (CI suite)

accepts a valid token for an active worker

Requirement: NFR-12 | Level: Unit | Priority: P1

Preconditions	Prepared isolated fixtures
Input / steps	Run named test with fixture data
Expected result	Named assertions succeed
Recorded actual result	Test suite passed in matched GitHub Actions CI; individual assertion transcript not archived.
Test file / evidence	backend/src/realtime/wkms-realtime.gateway.spec.ts

TC-WKMS-032 | PASS (CI suite)

rejects missing or empty tokens

Requirement: NFR-12 | Level: Unit | Priority: P1

Preconditions	Prepared isolated fixtures
Input / steps	Run named test with fixture data
Expected result	Named assertions succeed
Recorded actual result	Test suite passed in matched GitHub Actions CI; individual assertion transcript not archived.
Test file / evidence	backend/src/realtime/wkms-realtime.gateway.spec.ts

TC-WKMS-033 | PASS (CI suite)

refreshes shared data when the browser tab becomes visible

Requirement: NFR-03 | Level: Frontend Jest | Priority: P1

Preconditions	Prepared isolated fixtures
Input / steps	Run named test with fixture data
Expected result	Named assertions succeed
Recorded actual result	Test suite passed in matched GitHub Actions CI; individual assertion transcript not archived.
Test file / evidence	frontend/src/providers/task-data-provider.test.tsx

TC-WKMS-034 | NOT TESTED

Full realtime propagation latency meets SRS target under integrated load

Requirement: NFR-03 | Level: Performance / E2E | Priority: P1

Preconditions	Staging broker/socket + event timestamps
Input / steps	Publish real task and measure all browser updates
Expected result	All updates within specified latency
Recorded actual result	No verified complete execution or acceptance evidence at this commit.
Test file / evidence	backend/src/realtime/wkms-realtime.gateway.ts

TC-WKMS-035 | NOT TESTED

Installable worker PWA works in supported browsers

Requirement: WM-P01 | Level: PWA acceptance | Priority: P1

Preconditions	Service worker/manifest installation path
Input / steps	Install worker application
Expected result	PWA installation accepted
Recorded actual result	No verified complete execution or acceptance evidence at this commit.
Test file / evidence	frontend/src/app/layout.tsx

TC-WKMS-036 | NOT TESTED

Offline tasks cache and synchronize when connection returns

Requirement: WM-P02 | Level: Offline acceptance | Priority: P1

Preconditions	Offline persistence and sync design
Input / steps	Take offline, update task, reconnect
Expected result	Saved updates sync safely
Recorded actual result	No verified complete execution or acceptance evidence at this commit.
Test file / evidence	frontend/src/providers/task-data-provider.tsx

TC-WKMS-037 | NOT TESTED

Cloud HTTPS end-to-end deployment and recovery verified

Requirement: NFR-19 | Level: Staging acceptance | Priority: P1

Preconditions	Secure public staging deployment
Input / steps	Smoke test TLS, broker and DB failover
Expected result	Deployment and recovery documented
Recorded actual result	No verified complete execution or acceptance evidence at this commit.
Test file / evidence	README.md

8. Acceptance and evidence conditions

- Recorded PASS (CI suite) status reflects automated suite execution on the cited repository revision and is not upgraded to a manual UAT or deployed multi-subsystem staging sign-off.
- Suite coverage and selected register rows are distinct from raw automated test-assertion totals (which now exceed 4,700 passing automated checks across the organization).
- Obtain final tagged release-candidate evidence across all six subsystem repositories and shared services (api-gateway, websocket-service, infra), and record reviewer sign-off before university submission.
- Complete live cross-service staging verification for FDS-FOSS-KMS-WKMS-MAD event chains, live external PayHere/Brevo/printer hardware integrations, and the 100/300-user staging performance benchmarks.

- Keep raw test artifacts, synthetic fixtures, and repository links accessible without exposing secrets, connection strings, JWTs, or personal guest data.